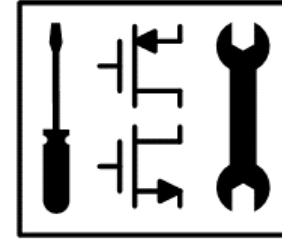


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Presents



The Design of Two-Stage Miller Op-Amp: The Final Verdict!

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adt.master-micro.com

Outline

□ The fundamentals

- Feedback and stability
- Frequency compensation and the optimal phase margin
- Why two-stage amplifier?

□ The two-stage amplifier

- Miller compensation
- The feedforward zero
- Design guidelines

□ The practice

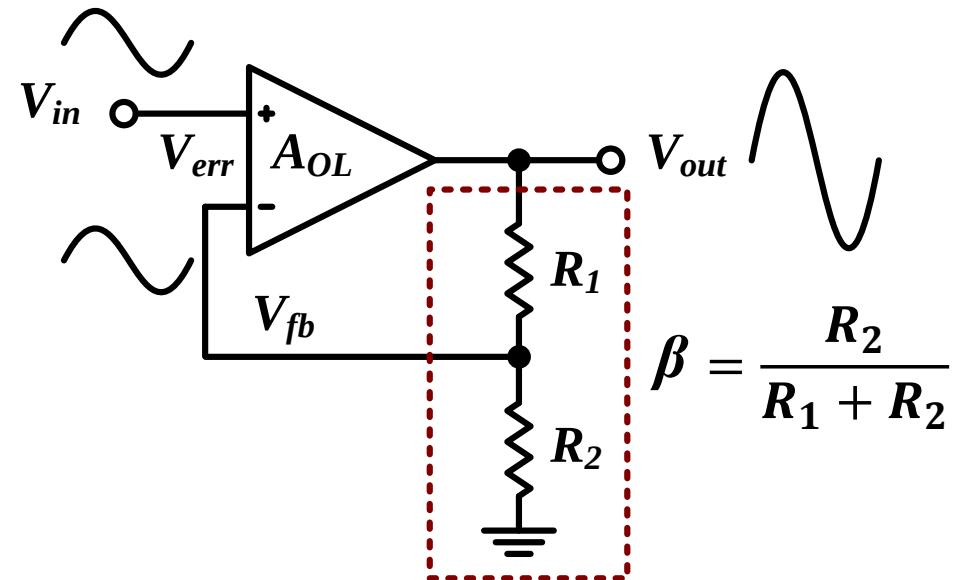
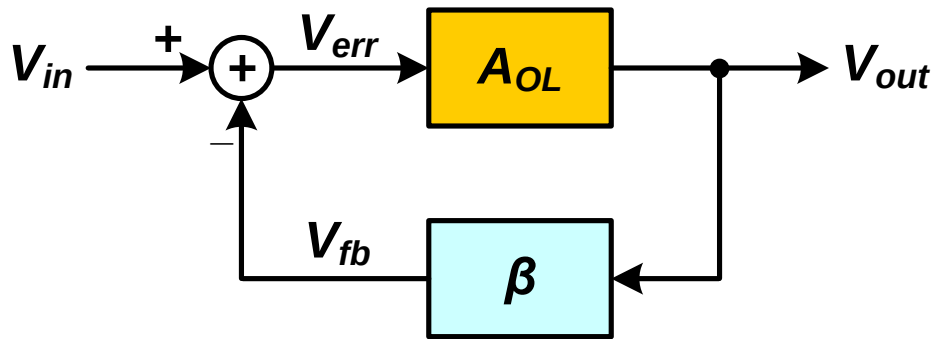
- Behavioral simulations
- Design two-stage amplifier in less than 2 minutes!

Why Do We Need High Gain Amplifier?

- ❑ Don't think about the output getting bigger but think about the input (V_{err}) getting smaller

$$A_{CL} = \frac{V_{out}}{V_{in}} = \frac{A_{OL}}{1 + \beta A_{OL}} \approx \frac{1}{\beta} = \frac{R_1 + R_2}{R_2} = 1 + \frac{R_1}{R_2}$$

$$\omega_{p,CL} = (1 + \beta A_{OL})\omega_{p,OL}$$

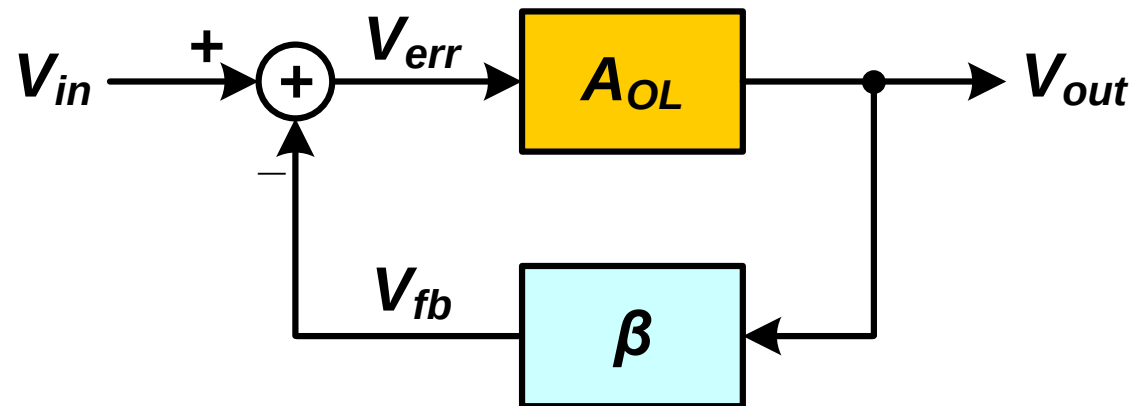


Stability of Negative Feedback System

$$A_{CL}(s) = \frac{V_{out}(s)}{V_{in}(s)} = \frac{A_{OL}(s)}{1 + \beta A_{OL}(s)}$$

□ Barkhausen's Oscillation Criteria

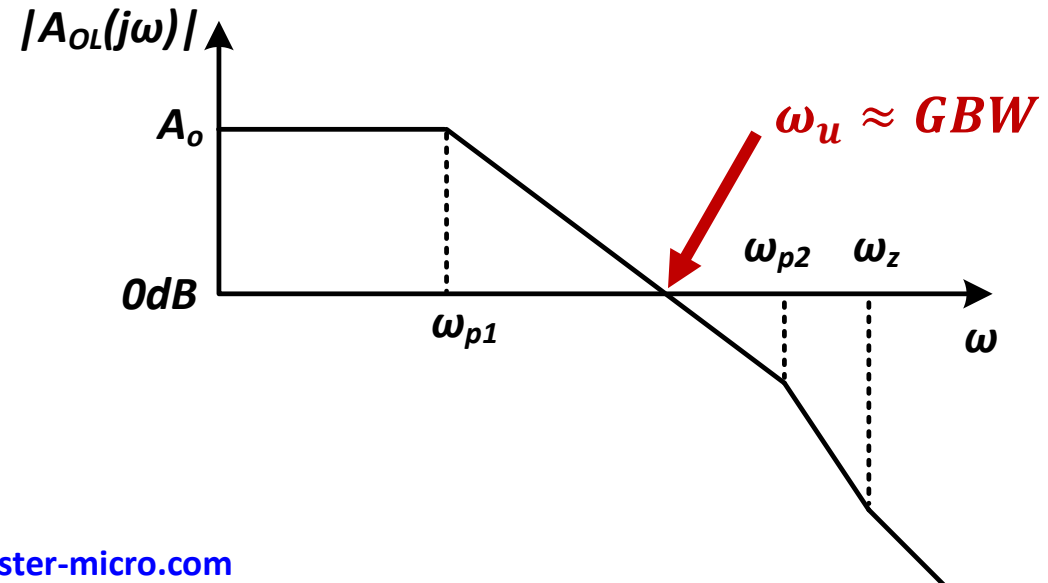
$$|\beta A_{OL}(s)| = 1$$
$$\angle \beta A_{OL}(s) = -180$$



IC Amplifier Frequency Response

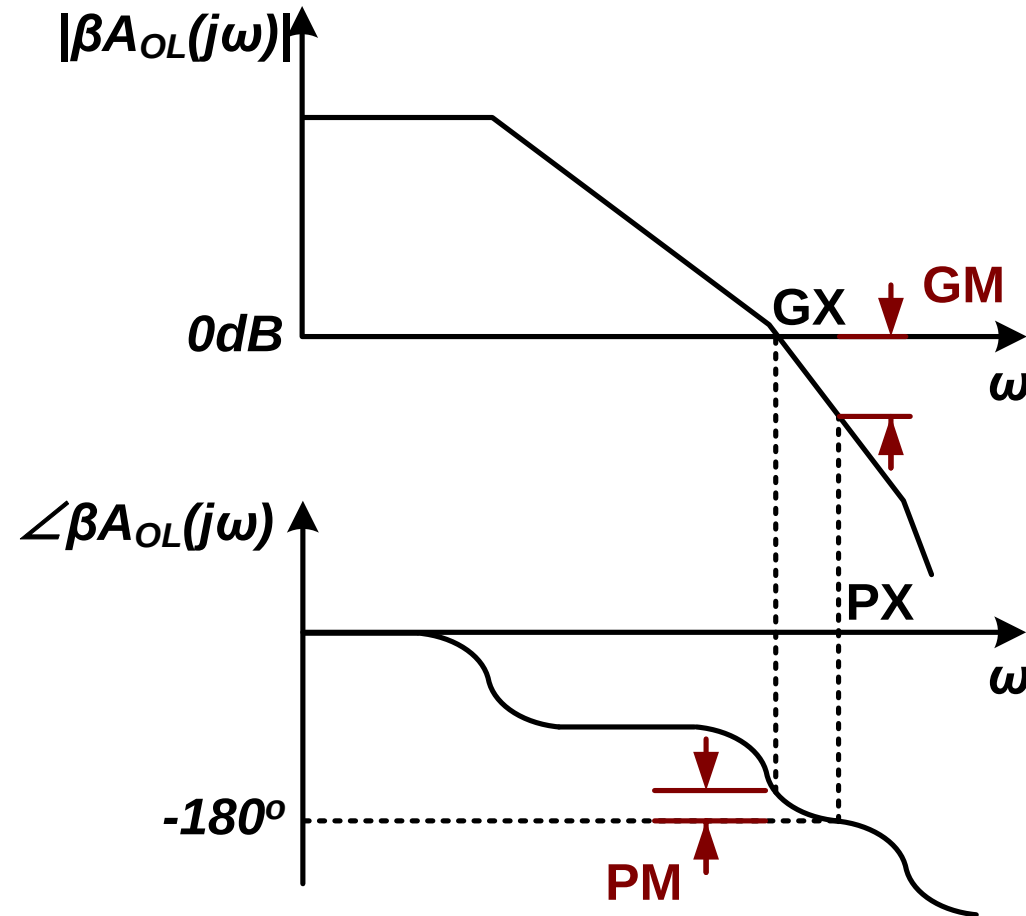
- ❑ A_o is the DC gain of the amplifier
- ❑ ω_{p1} is the dominant pole (ω_{pd}) $\approx$ 3dB bandwidth = $BW = \omega_{3dB}$
- ❑ ω_{p2} is the non-dominant pole (ω_{pnd})
- ❑ Unity gain frequency (UGF, ω_u) is the frequency at which gain is unity (1 = 0dB)
- ❑ Gain-Bandwidth Product (GBW) = $Gain \times BW \approx A_o \omega_{p1}$
- ❑ Usually, we design the amplifier such that ω_{p2} and $\omega_z > \omega_u$

$$A_{OL}(s) = \frac{A_o \left(1 + \frac{s}{\omega_z}\right)}{\left(1 + \frac{s}{\omega_{p1}}\right) \left(1 + \frac{s}{\omega_{p2}}\right)}$$

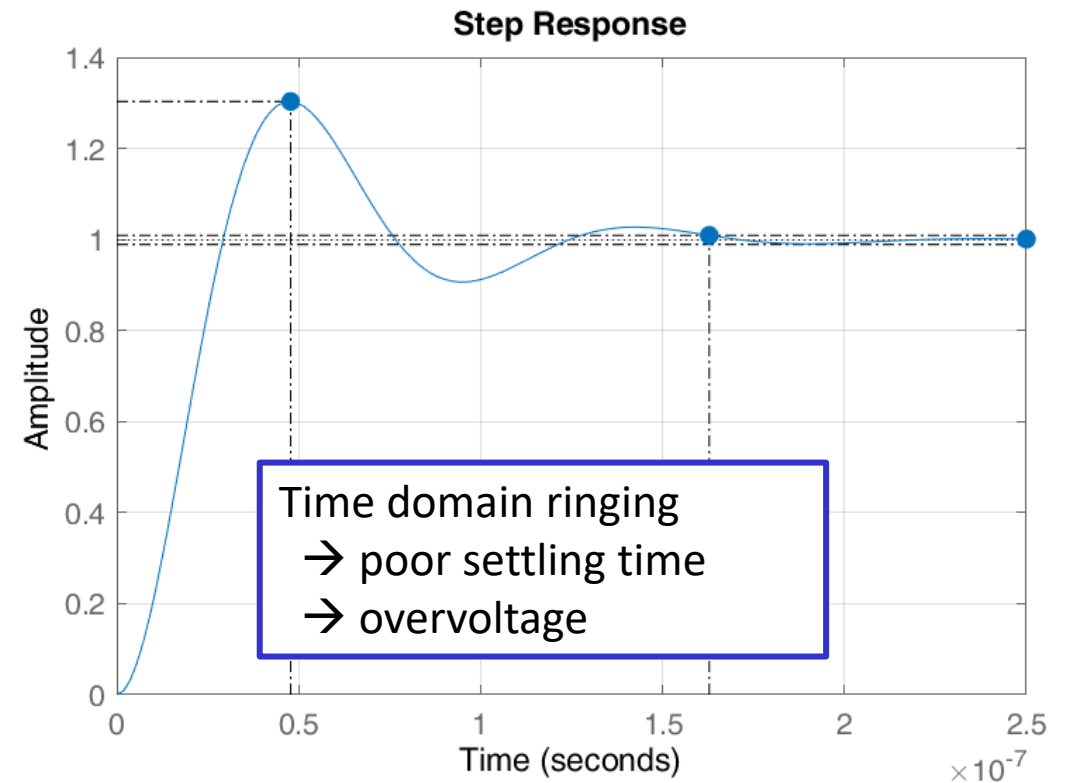
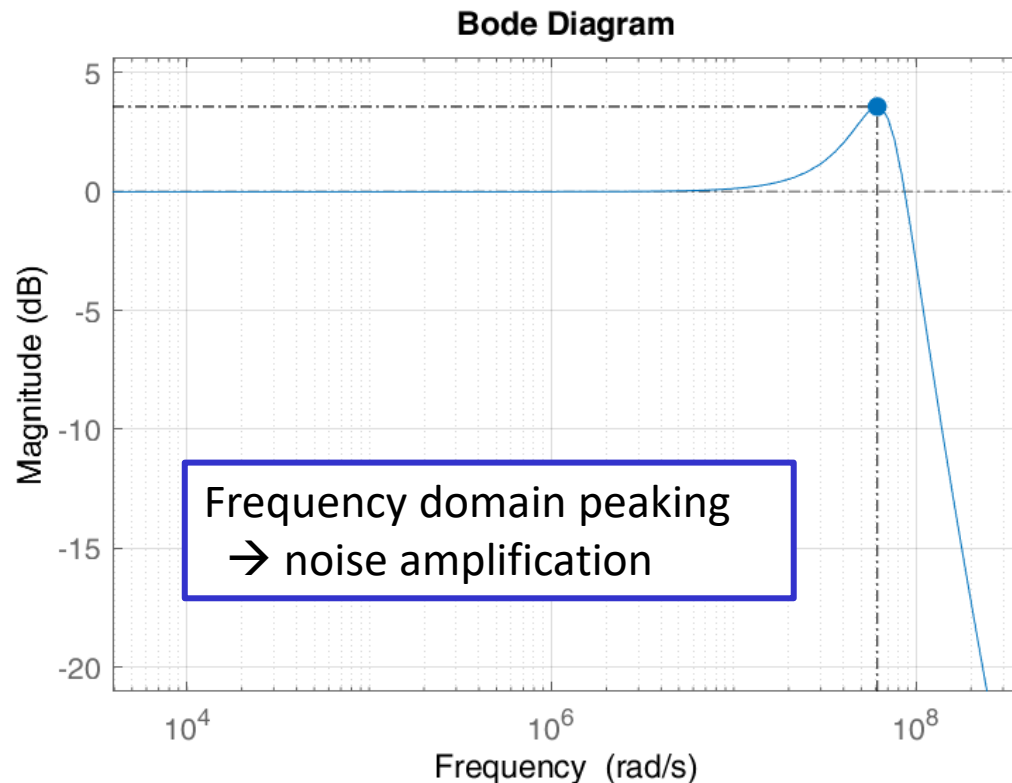
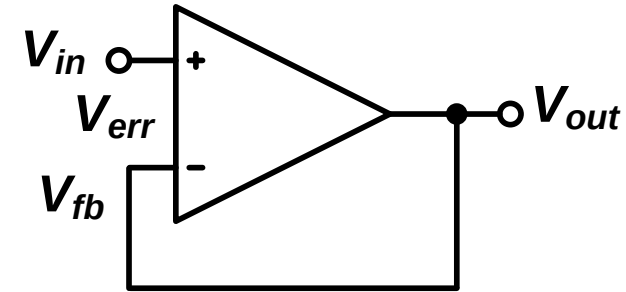
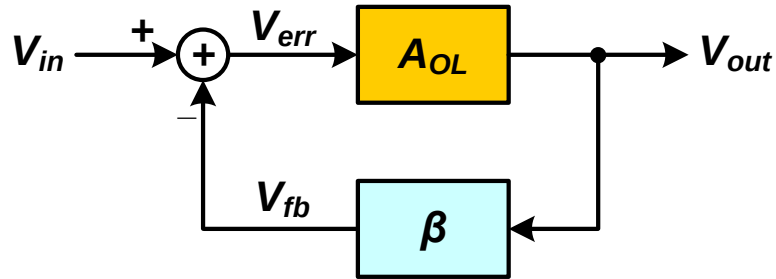


Phase Margin and Gain Margin

□ How much margin is needed?



Closed-Loop FD Peaking and TD Ringing (Overshoot)

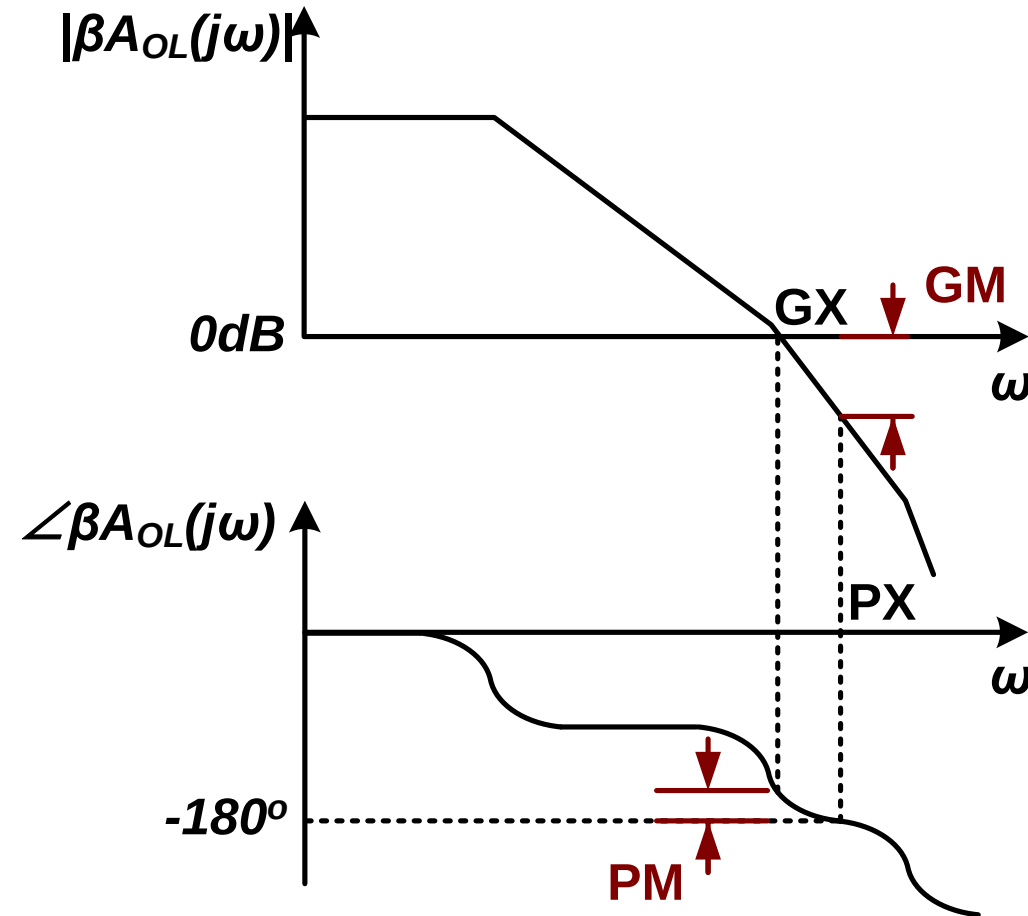


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Frequency Compensation

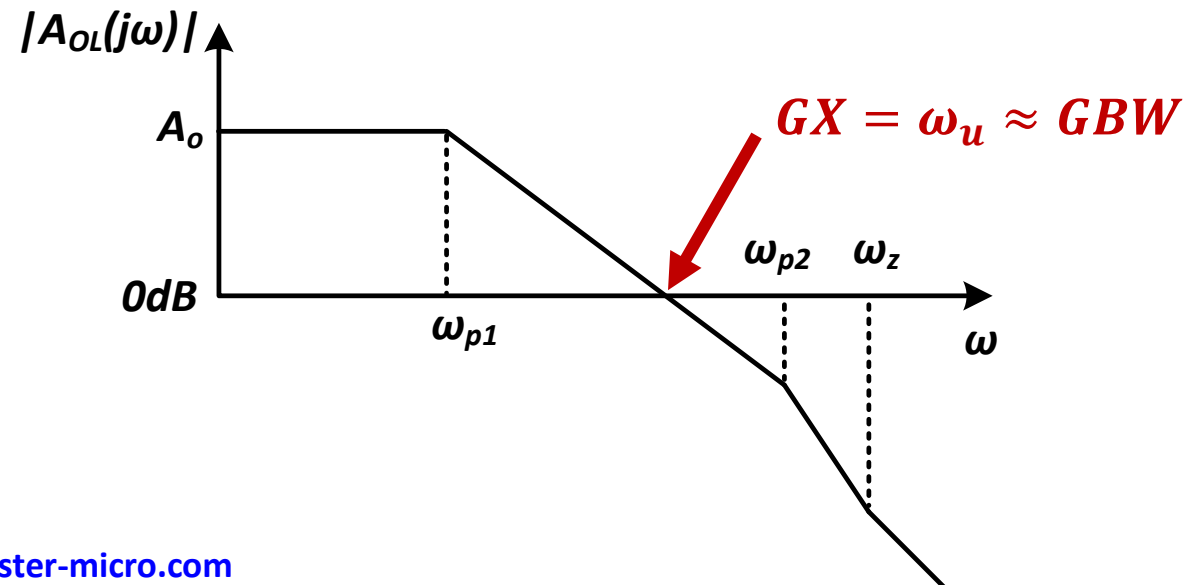
- Frequency compensation: the process of modifying the system to achieve a specific PM to control frequency domain peaking and time domain ringing.



How to Control the PM?

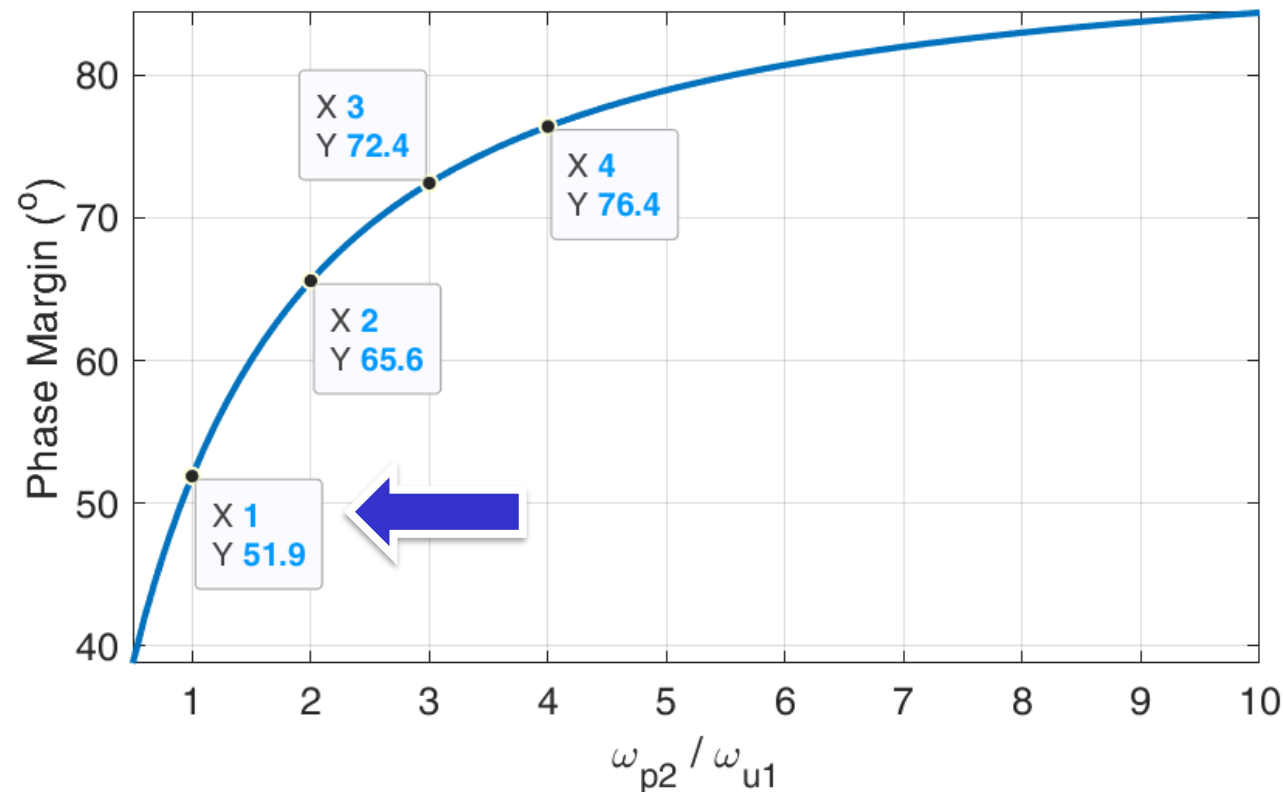
Control the ω_{p2}/ω_u ratio

$$A_{OL}(s) = \frac{A_o \left(1 + \frac{s}{\omega_z}\right)}{\left(1 + \frac{s}{\omega_{p1}}\right) \left(1 + \frac{s}{\omega_{p2}}\right)}$$



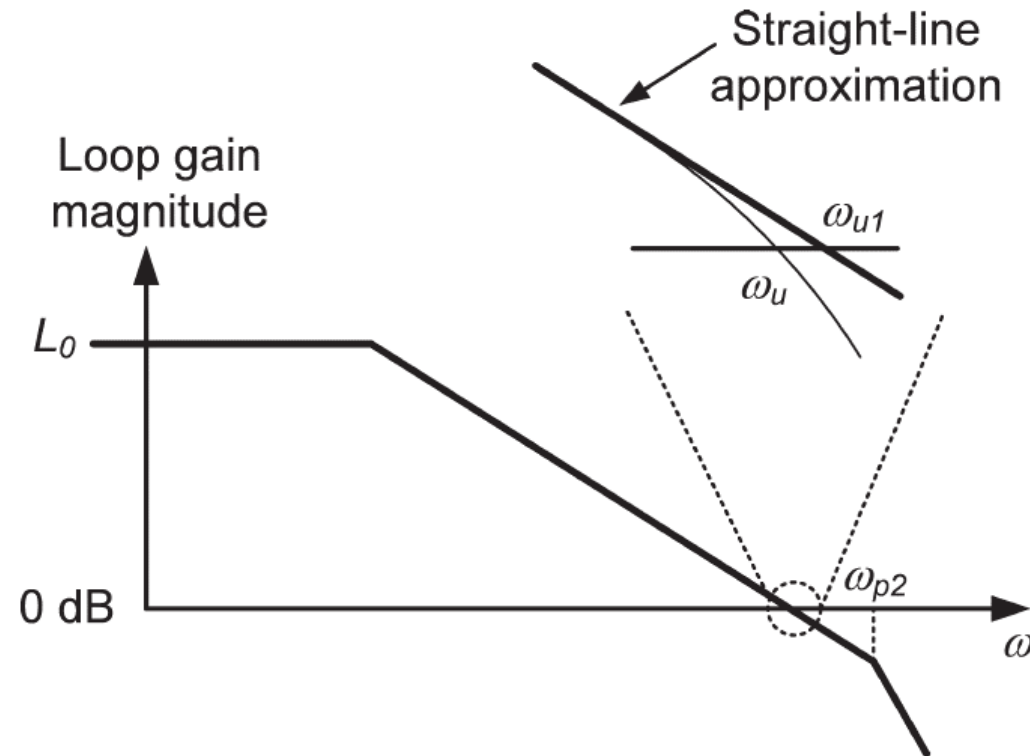
PM vs ω_{p2}/ω_u

- At $\omega_{p2}/\omega_u = 1$
 - We expect PM = 45°
 - But the actual PM is higher?!



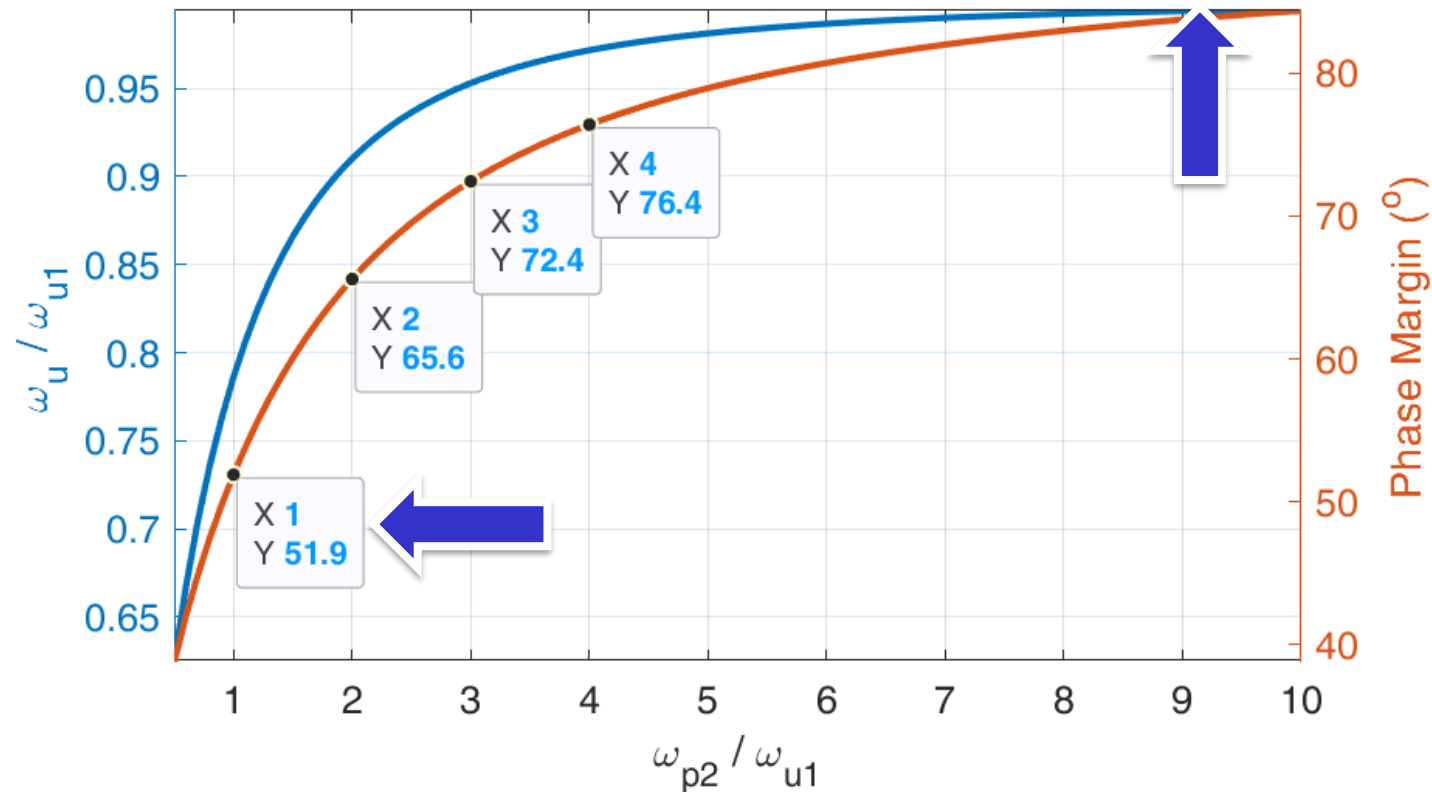
Asymptotic vs Actual UGF

- ❑ For $\omega < \omega_u$ the Bode plot is similar to a 1st order system
- ❑ For a true 1st order system: $\text{GBW} = \text{UGF} = \omega_u = \omega_{u1}$
- ❑ But the 2nd pole causes some bending below the asymptote
 - Actual ω_u is slightly smaller than ω_{u1} : $\text{UGF} < \text{GBW}$



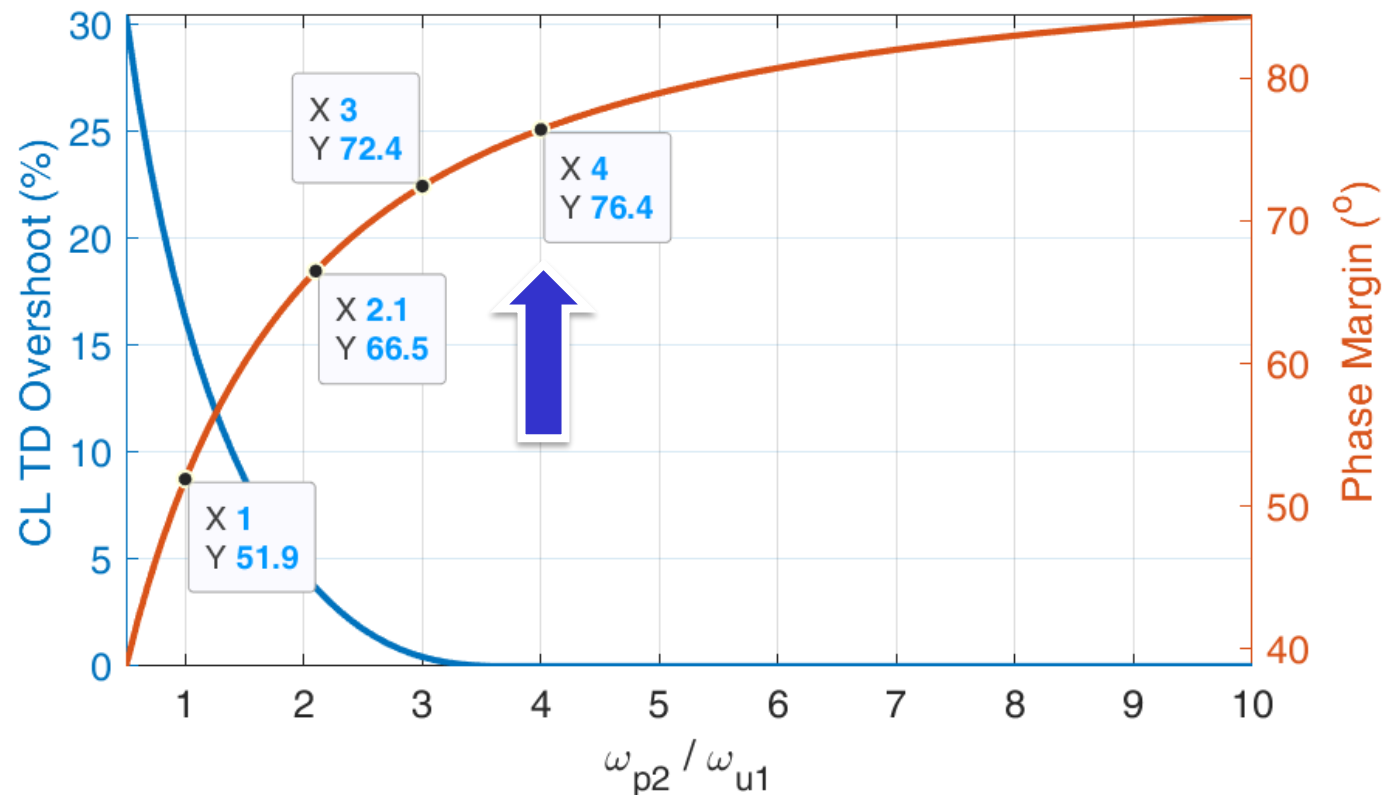
Asymptotic vs Actual UGF

- ❑ As ω_{p2}/ω_{u1} gets smaller the deviation is larger: $\omega_u = UGF = GX < \omega_{u1} = GBW$
- ❑ At $\omega_{p2}/\omega_{u1} \rightarrow \infty$: $\omega_u \approx \omega_{u1}$



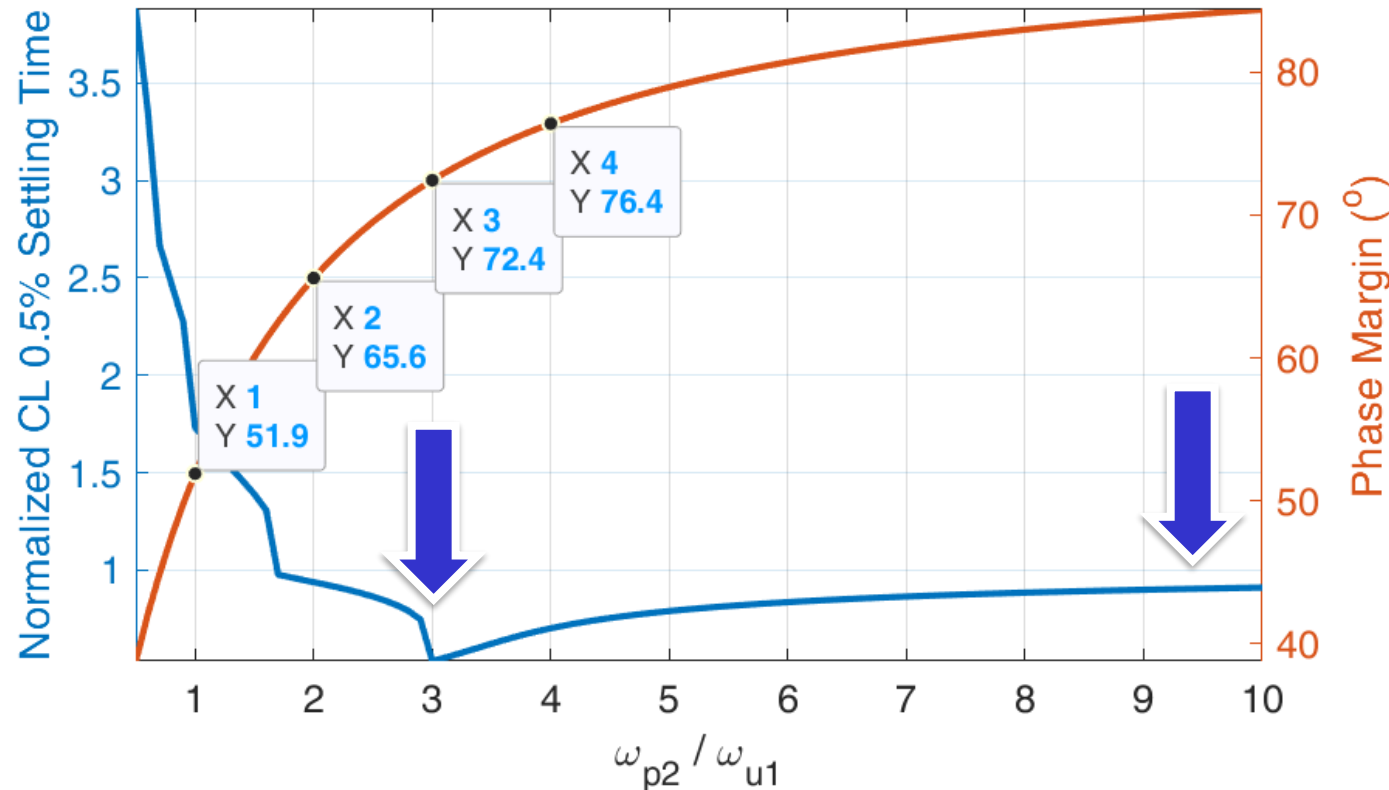
Optimal Phase Margin: Transient Response

- ❑ Closed-loop fastest settling without overshoot occurs at $\zeta = 1$ ($Q = \frac{1}{2\zeta} = 0.5$)
 - Critically damped system
 - $\omega_{p2}/\omega_u = 4$ and $PM \approx 76^\circ$



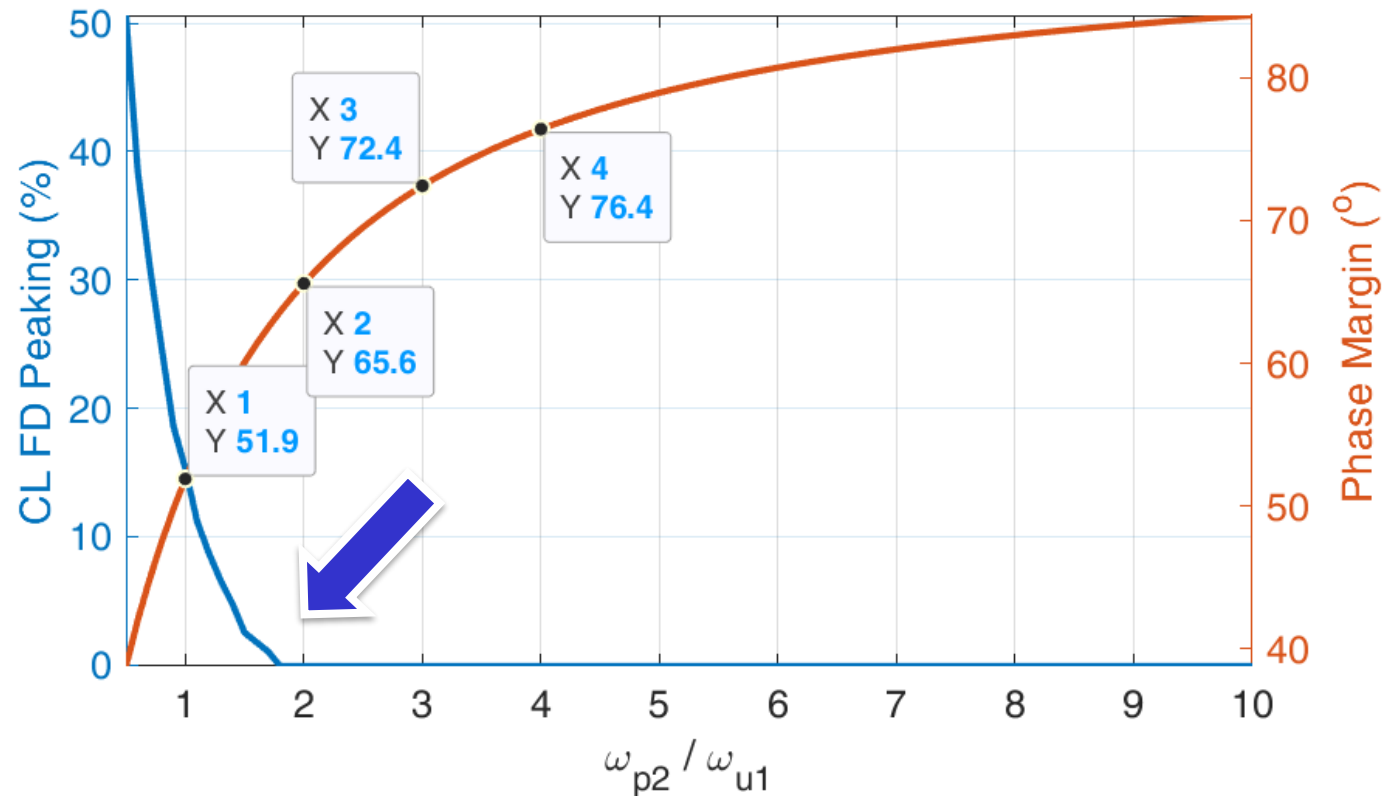
Optimal Phase Margin: Transient Response

- ❑ But faster settling can be achieved if we tolerate some slight overshoot
 - $\omega_{p2}/\omega_u = 3$ and $PM \approx 72^\circ$
- ❑ The settling of 2nd order system with optimum PM is faster than 1st order system



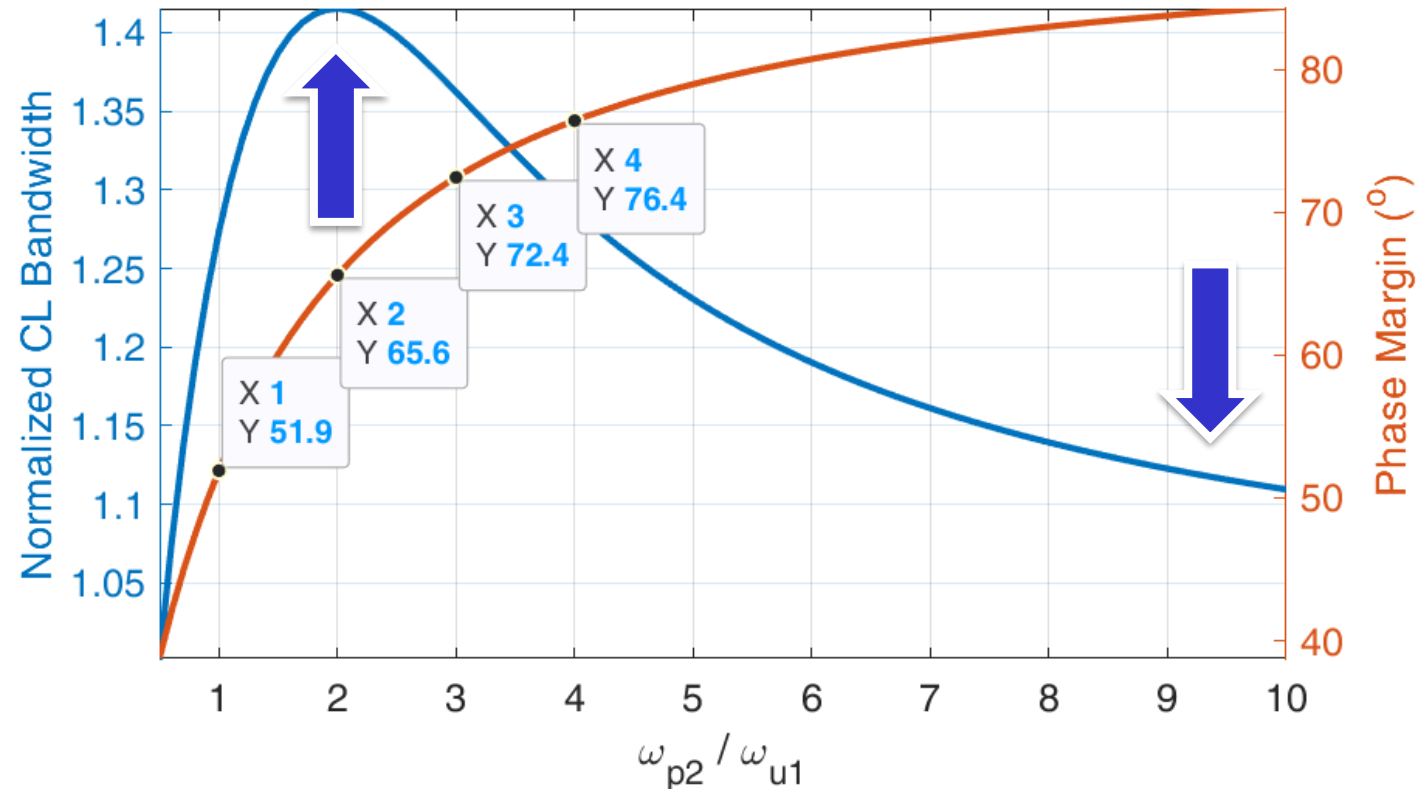
Optimal Phase Margin: Frequency Response

- ❑ Maximum closed-loop BW without peaking occurs at $\zeta = Q = 0.707$
 - Maximally flat response
 - $\omega_{p2}/\omega_u = 2$ and $PM \approx 66^\circ$
- ❑ But the system is underdamped: overshoot exists in the transient response



Optimal Phase Margin: Frequency Response

- ❑ Maximum closed-loop BW without peaking occurs at $\zeta = Q = 0.707$
 - Maximally flat response
 - $\omega_{p2} = 2\omega_{u1}$ and $PM \approx 66^\circ$
- ❑ But the system is underdamped: overshoot exists in the transient response



Optimal Phase Margin: Summary

PM (°)	ω_{p2}/ω_{u1}	0.5% Settling	Overshoot (%)	Bandwidth	Notes
52	1	1.73x	16.3	1.27x	15.3% peaking
66	2	0.94x	4.32	1.42x	Maximally flat response No peaking Some overshoot
72	3	0.53x	0.43	1.36x	Fastest settling No peaking Very slight overshoot
76	4	0.7x	No	1.29x	Critically damped response No peaking No overshoot

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Poles by Intuition

- ❑ Each node is associated with a pole

$$\omega_p = \frac{1}{R_{th}C}$$

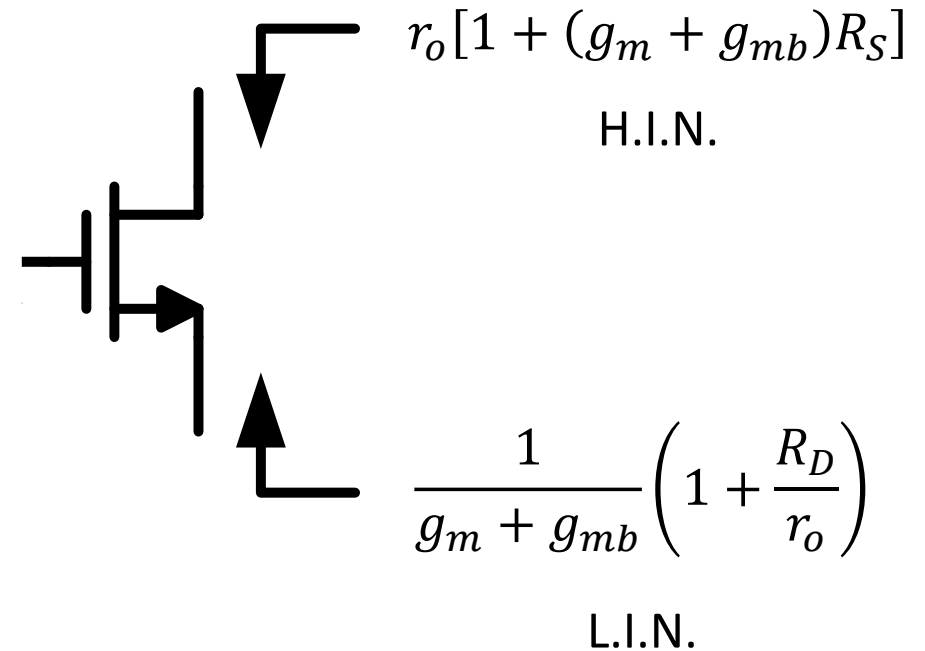
- ❑ High impedance nodes (H.I.N.s)

- Give dominant poles
- Also give gain

$$A_v = G_m R_{th}$$

- ❑ Low impedance nodes (L.I.N.s)

- Give non-dominant poles



Single-Stage Amplifier

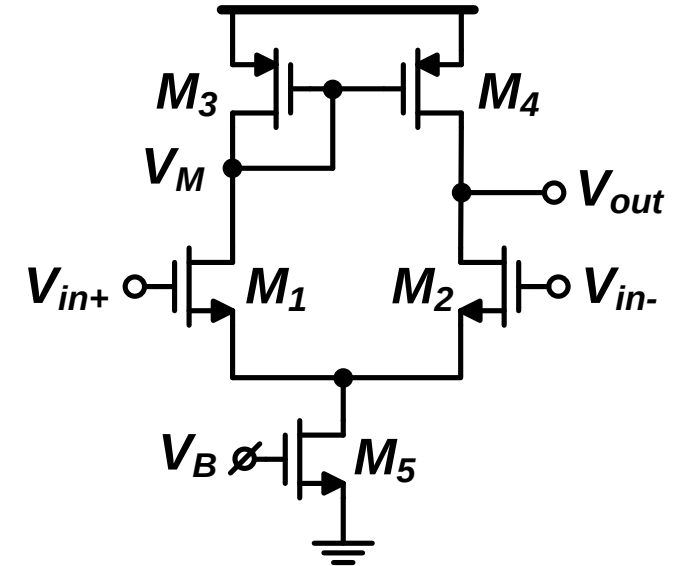
❑ Single H.I.N.

$$A_v \approx \frac{g_m r_o}{2} = \frac{1}{2} \cdot \frac{2I_D}{V^*} \cdot \frac{V_A}{I_D} = \frac{V_A}{V^*}$$

❑ The gain is limited.

❑ The intrinsic gain ($g_m r_o$) declines with technology scaling.

- $V_A \propto L$

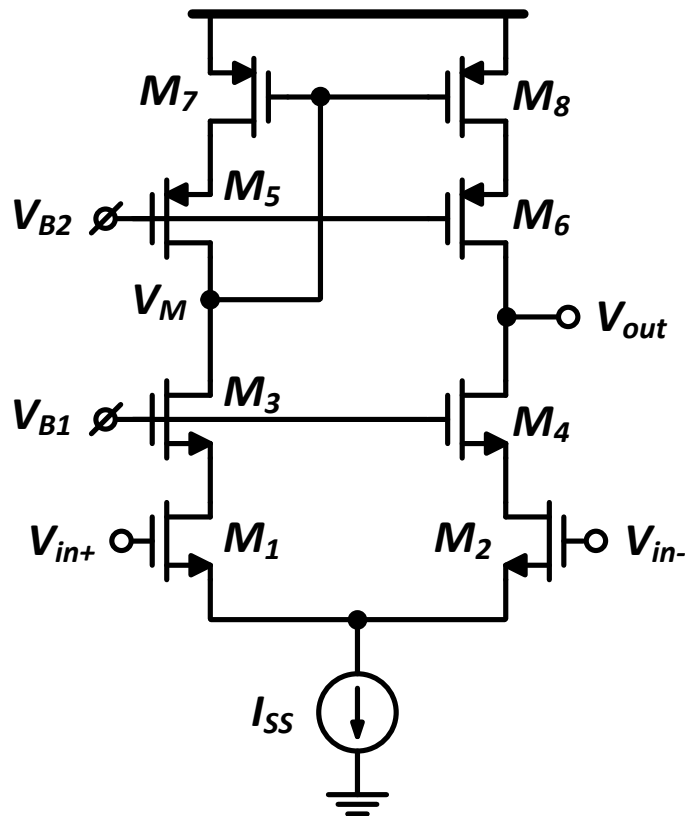


Cascode Amplifier

Telescopic cascode

CS + CG

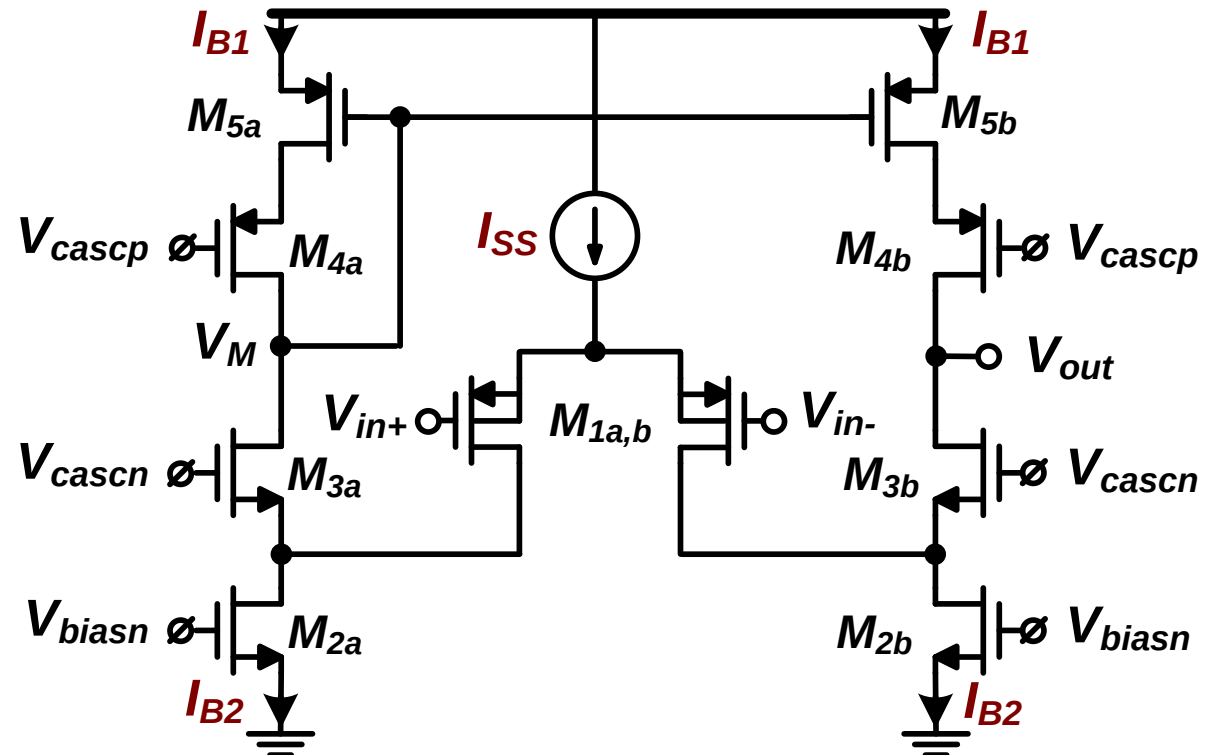
Both NMOS or both PMOS



Folded cascode

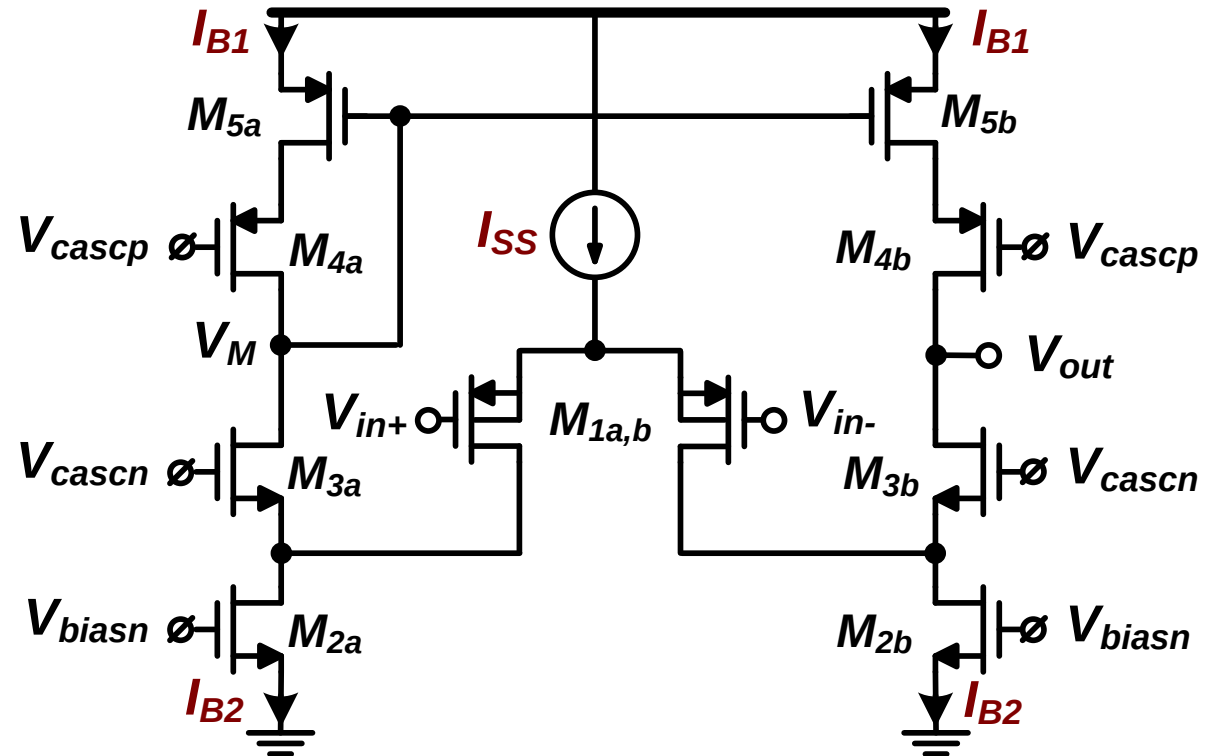
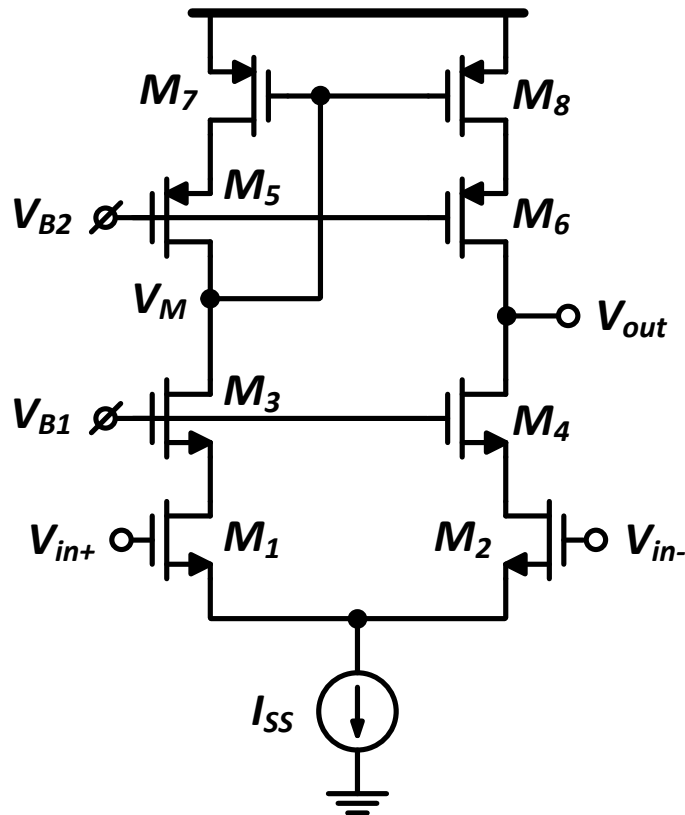
CS + CG

NMOS-PMOS combination



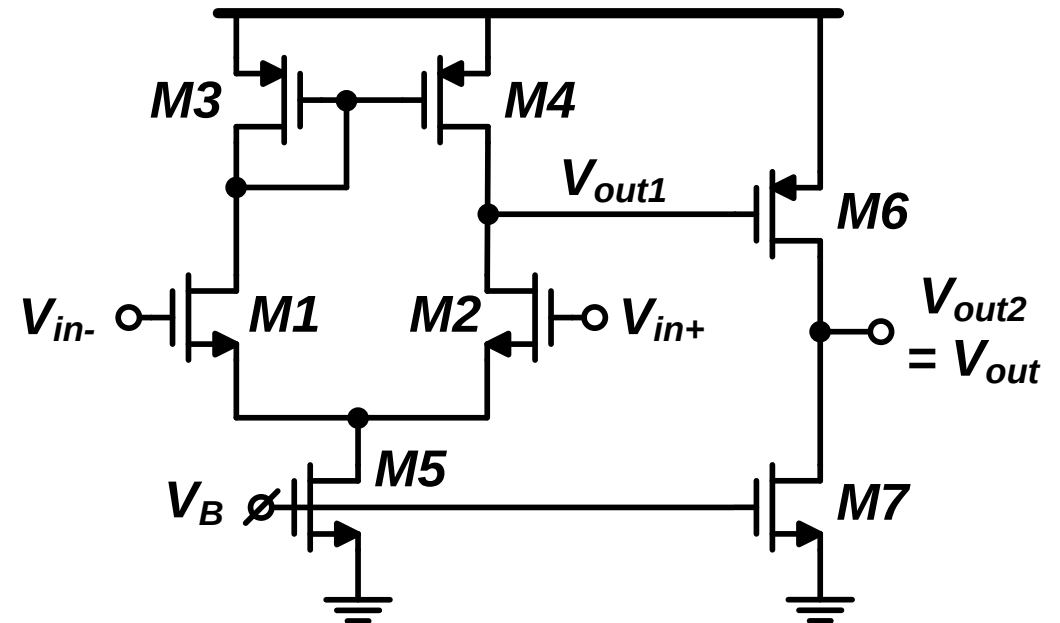
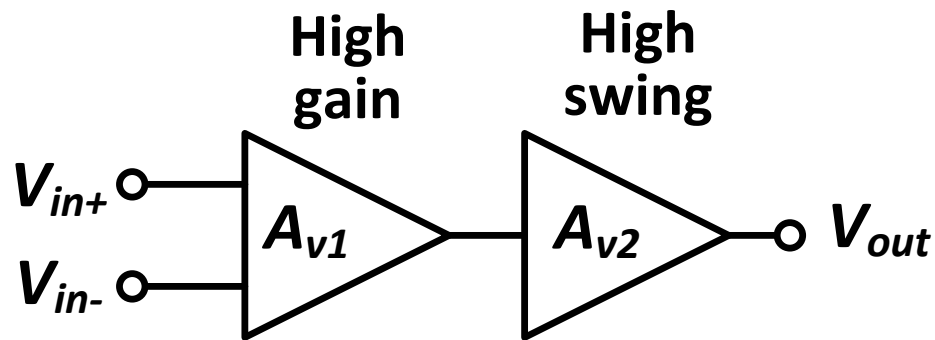
Cascode Amplifier

- ❑ Both have single H.I.N.
- ❑ Can give high gain $\sim (g_m r_o)^2$
- ❑ But suffer from limited headroom (signal swing)



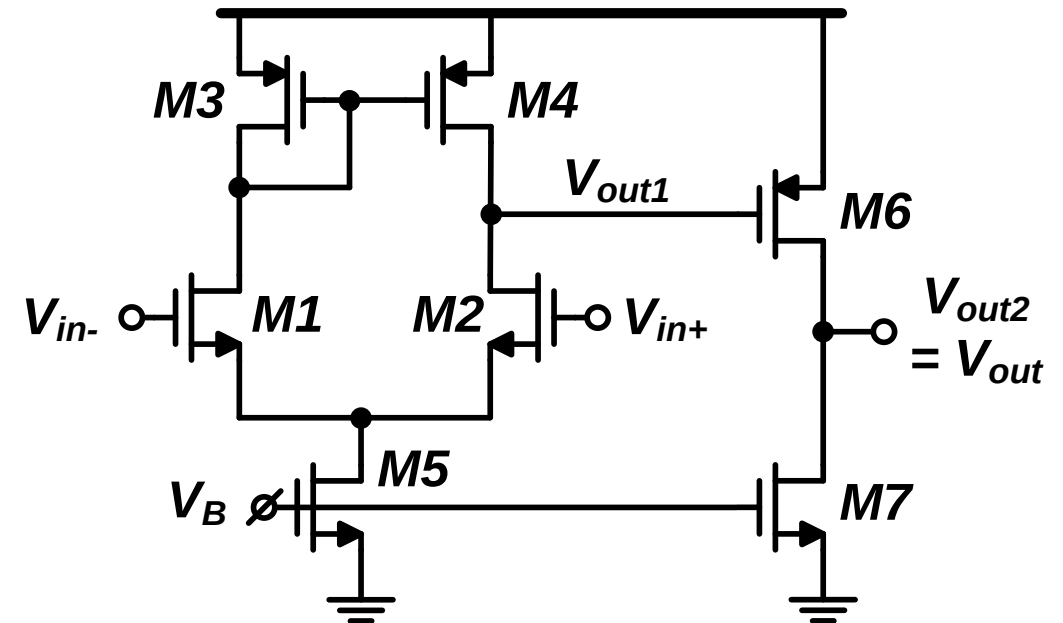
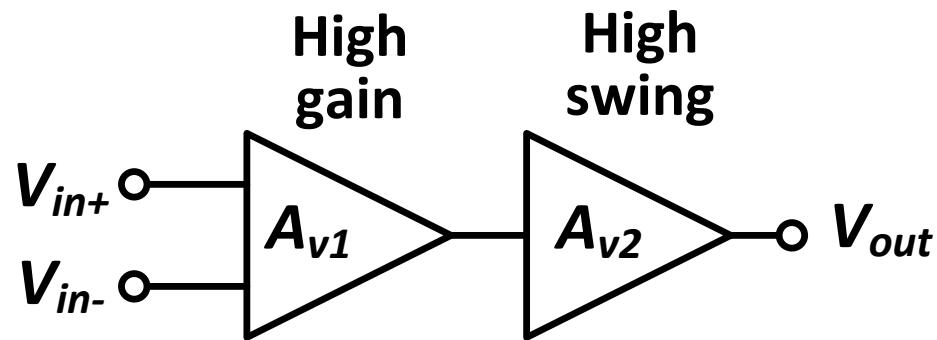
Two-Stage Amplifier

- ❑ Isolates the gain and swing requirements
 - Don't think about the output getting bigger but think about the input getting smaller
- ❑ First stage can be 5T-OTA or cascode.
- ❑ Second stage is typically a simple common-source.
 - Allows maximum output swing.



Two-Stage Amplifier

- ❑ But two H.I.N.s
 - Complicates the stability requirements.
- ❑ And more power consumption.

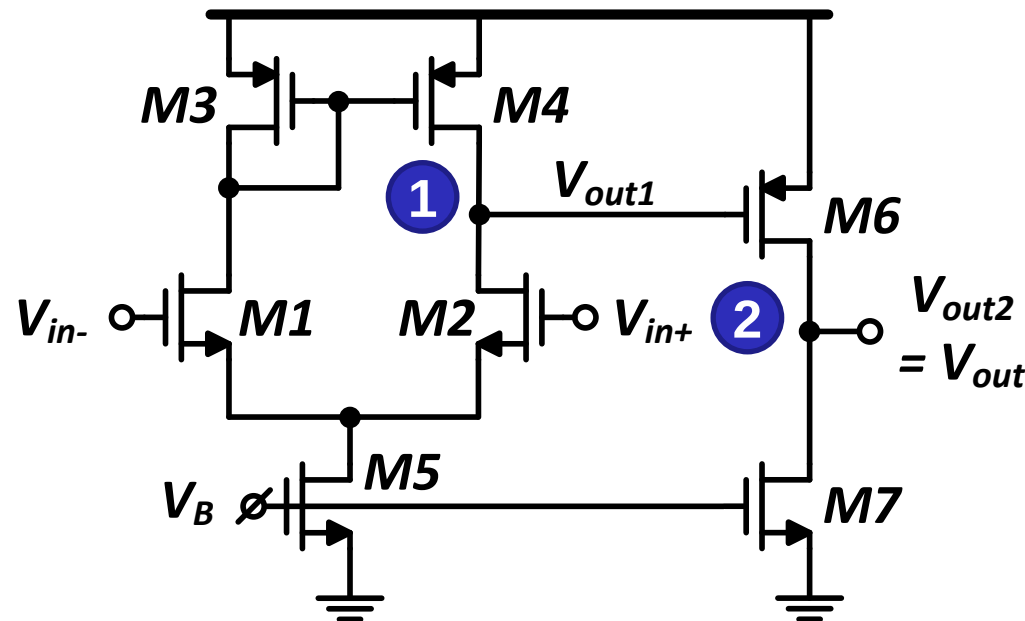


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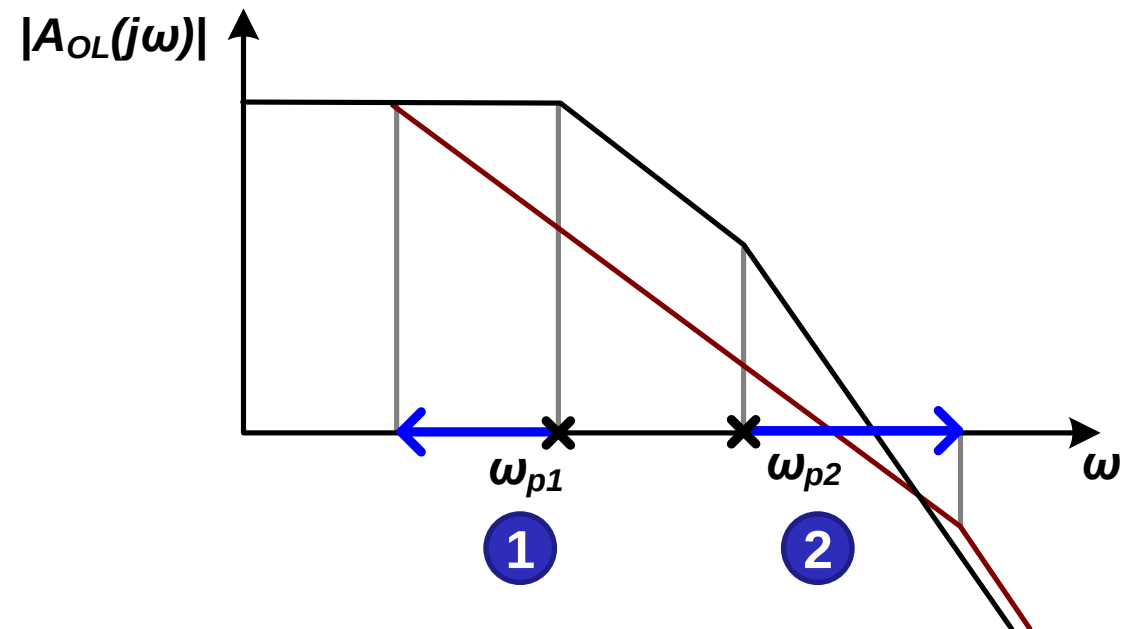
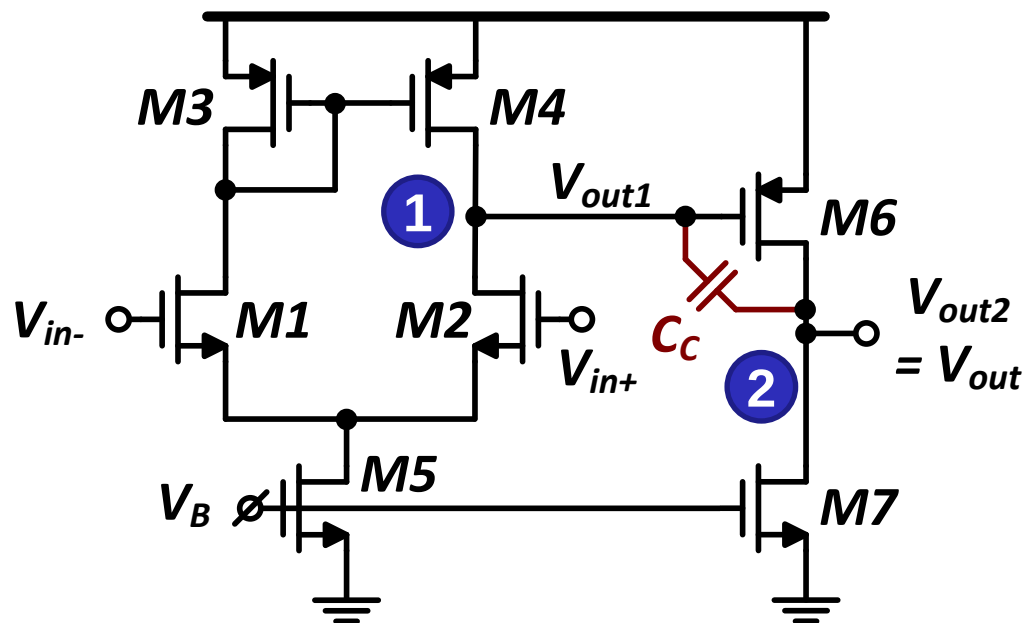
Internal and Output Poles

- ❑ Two gain stages → Two H.I.N.s
 - Two dominant poles!
- ❑ Internal pole: $\omega_{p1} = \frac{1}{R_{out1}C_1}$
- ❑ Output pole: $\omega_{p2} = \frac{1}{R_{out2}C_2}$



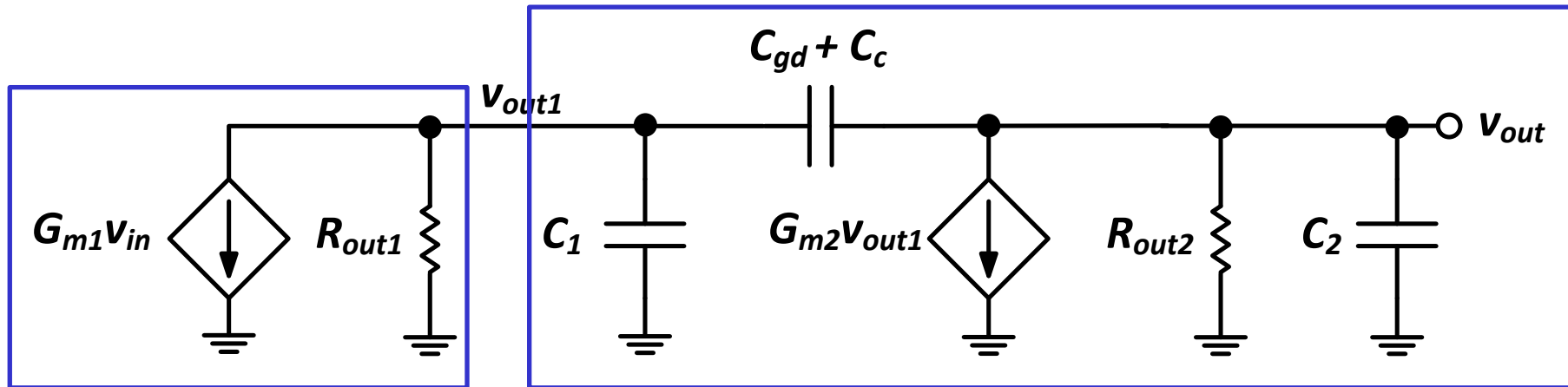
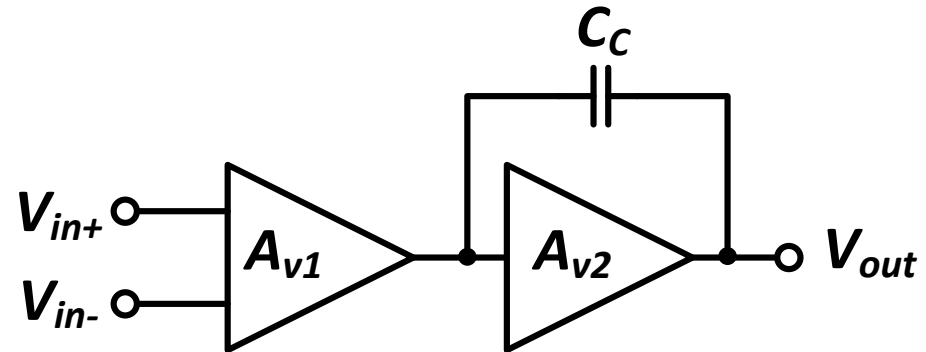
Miller Compensation

- ❑ Exploit Miller capacitance multiplication
- ❑ Pole splitting
 - Push ω_{p1} inwards $\rightarrow$ push GX (ω_u) inwards
 - Push ω_{p2} outwards $\rightarrow$ push PX outwards



High Frequency Analysis

- ❑ Surprisingly, exact analysis gives a quite complex expression
 - See [Johns & Martin 2012] [Razavi 2017] or use a symbolic solver (e.g., SLiCAP)!



The Dominant Pole Approximation

- ❑ Surprisingly, exact analysis gives a quite complex expression

$$A_{OL}(s) \approx \frac{A_o}{1 + b_1 s + b_2 s^2}$$

- ❑ If the dominant pole approximation is applied on the exact transfer function

$$\omega_{p1} \approx \frac{1}{b_1} = \frac{1}{R_{out1}[C_C(1 + G_{m2}R_{out2}) + C_1] + R_{out2}(C_C + C_2)}$$

- Same result as OCTC method (both based on same approximation)

- ❑ Additionally, the dominant pole approx gives an expression for the non-dominant pole

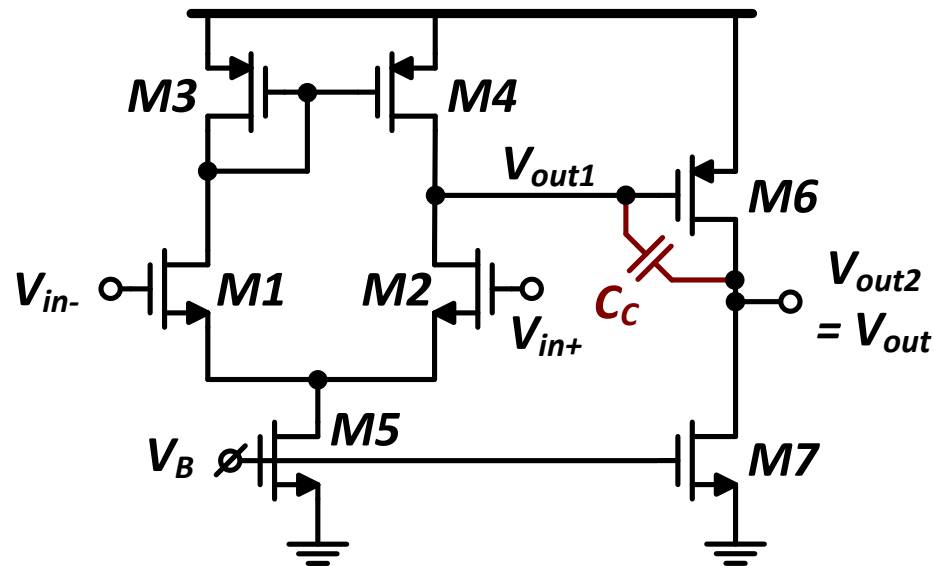
$$\omega_{p2} \approx \frac{1}{b_2 \omega_{p1}} = \frac{b_1}{b_2} = \frac{G_{m2}C_C}{C_C(C_1 + C_2) + C_1C_2}$$

Intuitive Analysis

- Poles can be derived intuitively without complicated analysis!

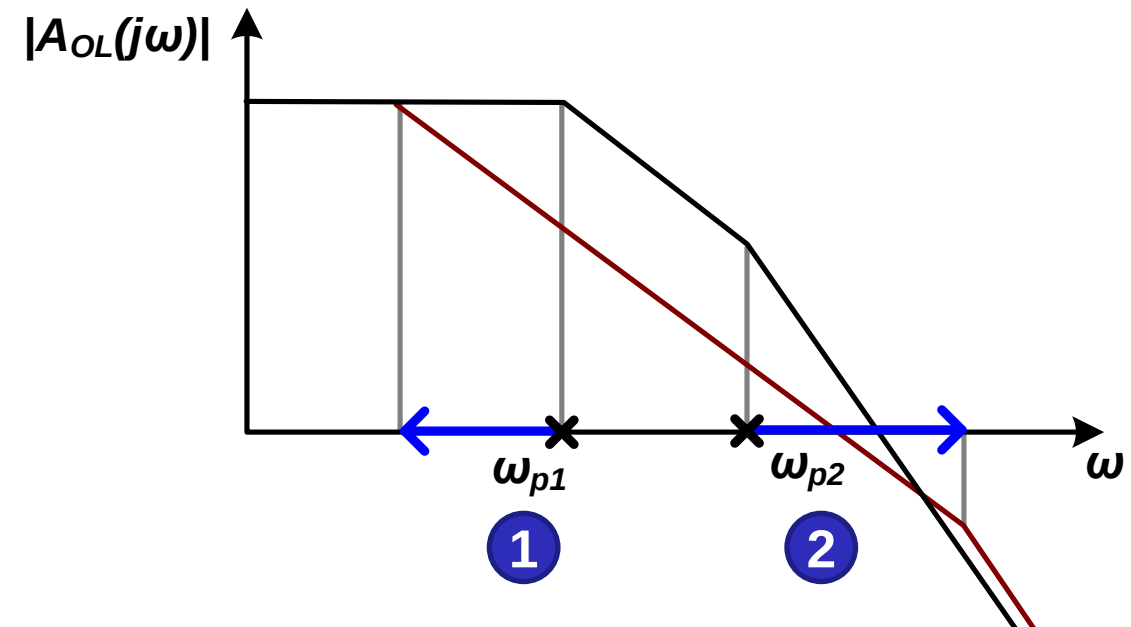
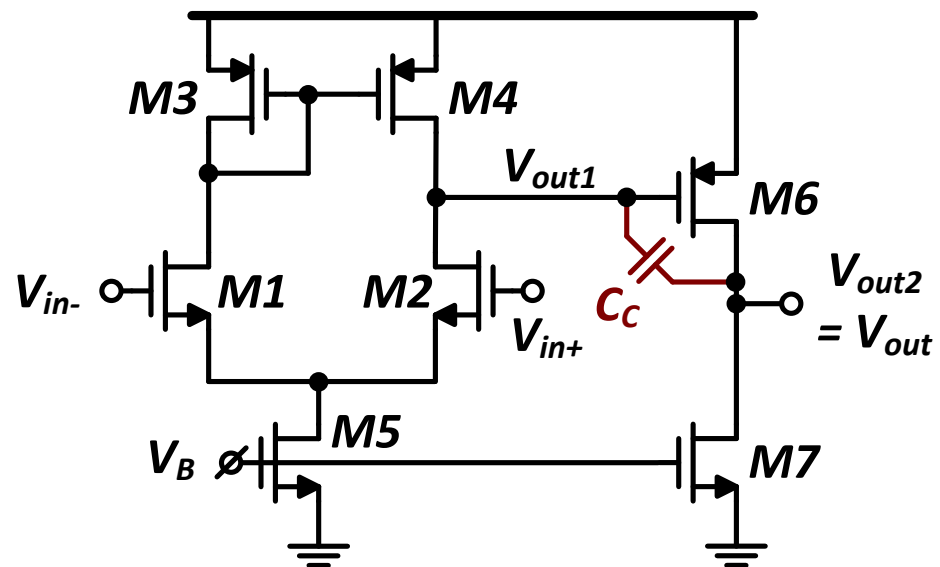
$$\omega_{p1} \approx \frac{1}{R_{out1}[(G_{m2}R_{out2})C_C + C_1] + R_{out2}(C_C + C_2)} \approx \frac{1}{R_{out1}(G_{m2}R_{out2})C_C}$$

$$\omega_{p2} \approx \frac{G_{m2}C_C}{C_C(C_1 + C_2) + C_1C_2} \approx \frac{G_{m2}}{C_1 + C_2}$$



Miller Compensation: Pole Splitting

	ω_{p1}	ω_{p2}
Before	$\frac{1}{R_{out1}C_1}$	$\frac{1}{R_{out2}C_2}$
After	$\frac{1}{R_{out1}(G_{m2}R_{out2})C_C}$	$\frac{G_{m2}}{C_1 + C_2}$



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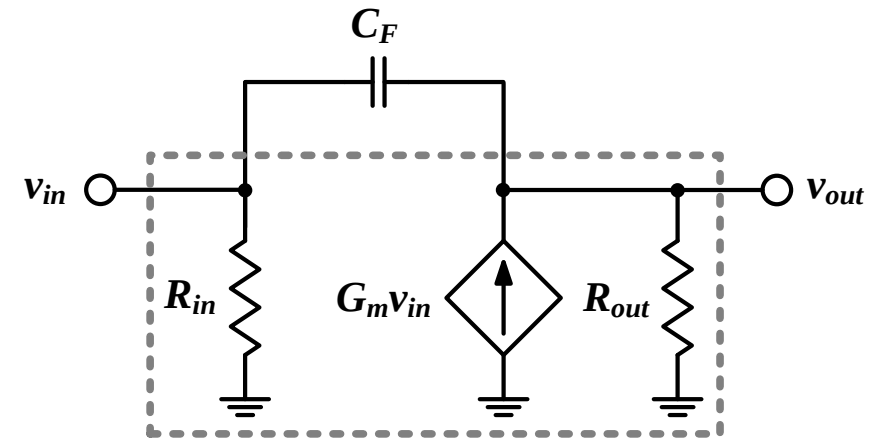
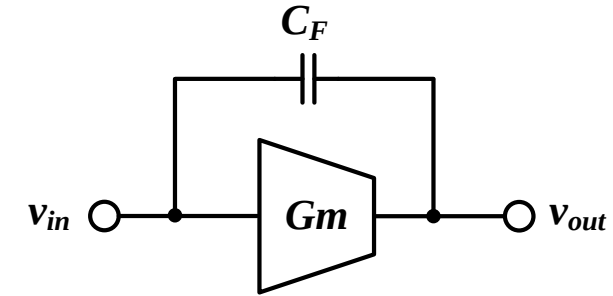
The Feedforward Zero

$$v_{out} = 0 \rightarrow i_{out} = 0$$

$$v_{in} s C_F + G_m v_{in} = 0$$

$$s_z = -\frac{G_m}{C_F}$$

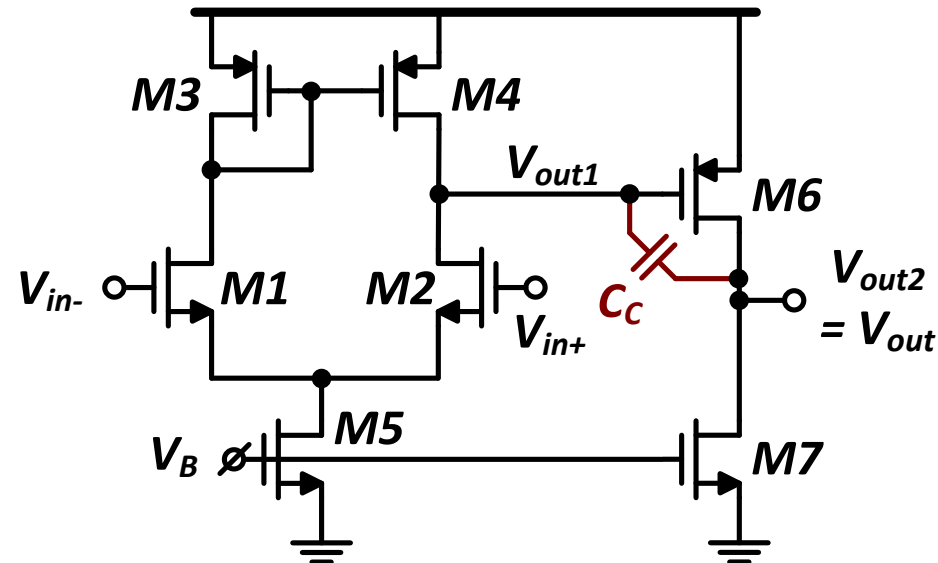
- ❑ LHP zero if G_m is +ve (e.g., CD)
- ❑ RHP zero if G_m is -ve (e.g., CS)



The RHP Zero

- ❑ RHP zero is bad for both magnitude and phase
 - Pushes GX outwards and pushes PX inwards
 - Increasing C_C may hurt stability!

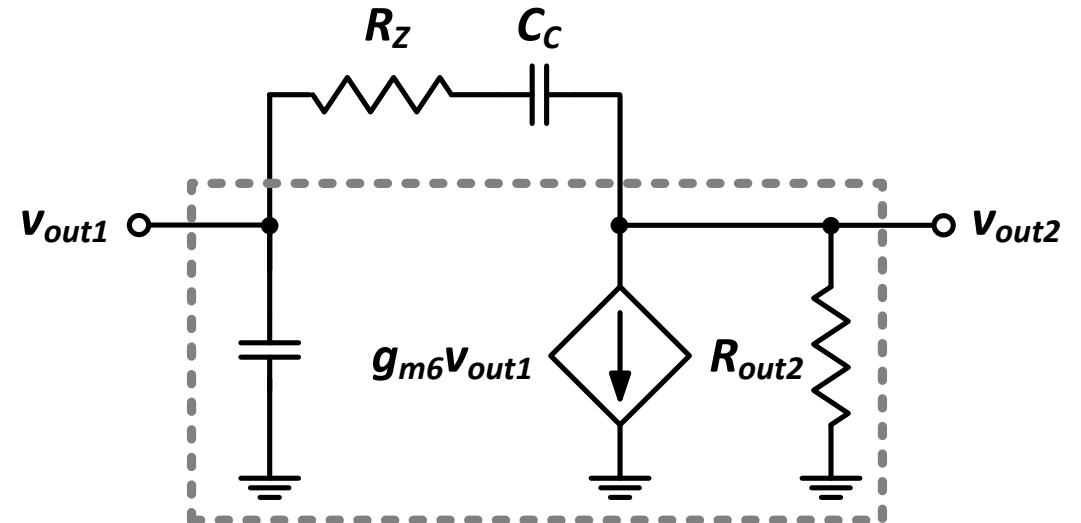
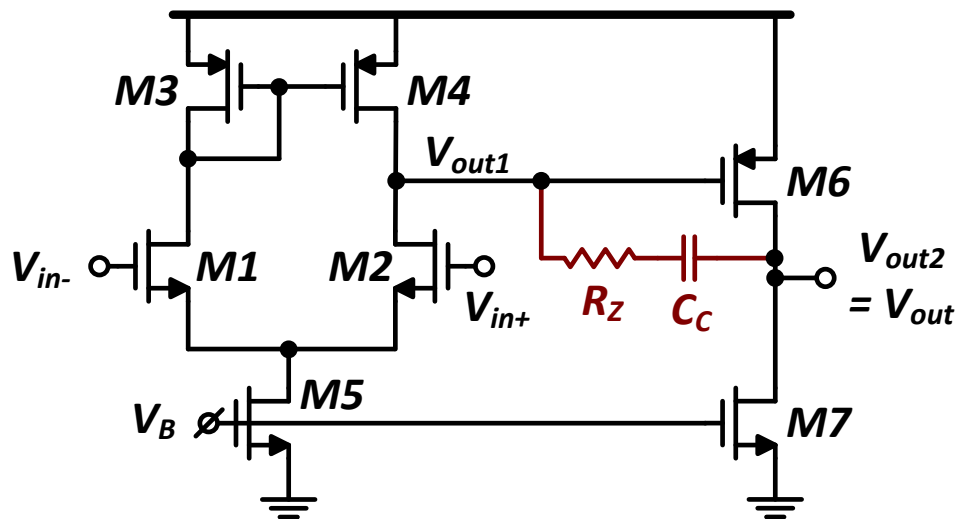
$$\omega_z = \frac{g_{m6}}{C_C + C_{gd}} \approx \frac{G_{m2}}{C_C}$$



Handling the RHP Zero

- Add a resistance to control the value of the zero (other tricks exist)

$$\frac{v_{out1}}{R_Z + \frac{1}{s_Z C_C}} = g_{m6} v_{out1}$$
$$s_Z = \frac{1}{C_C \left(\frac{1}{g_{m6}} - R_Z \right)}$$

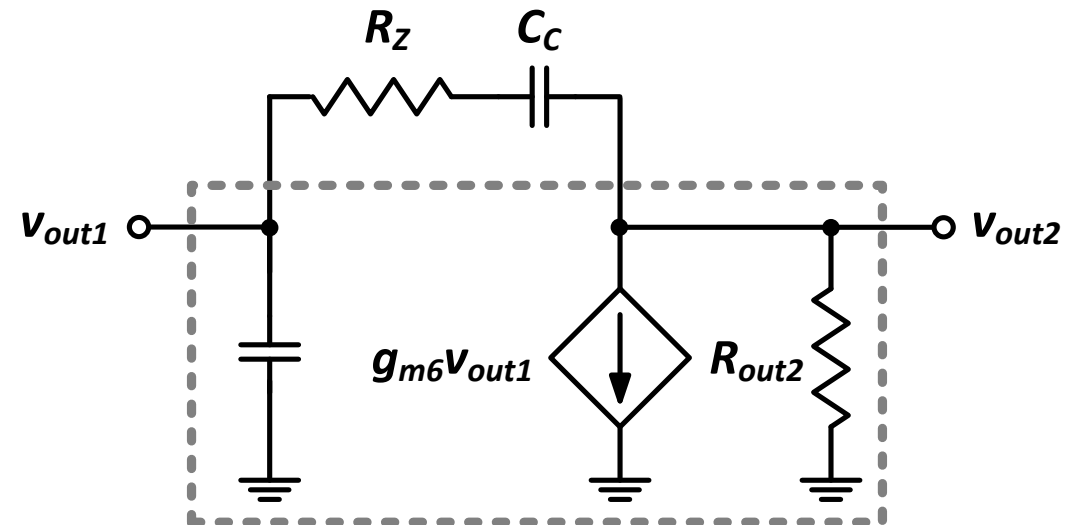
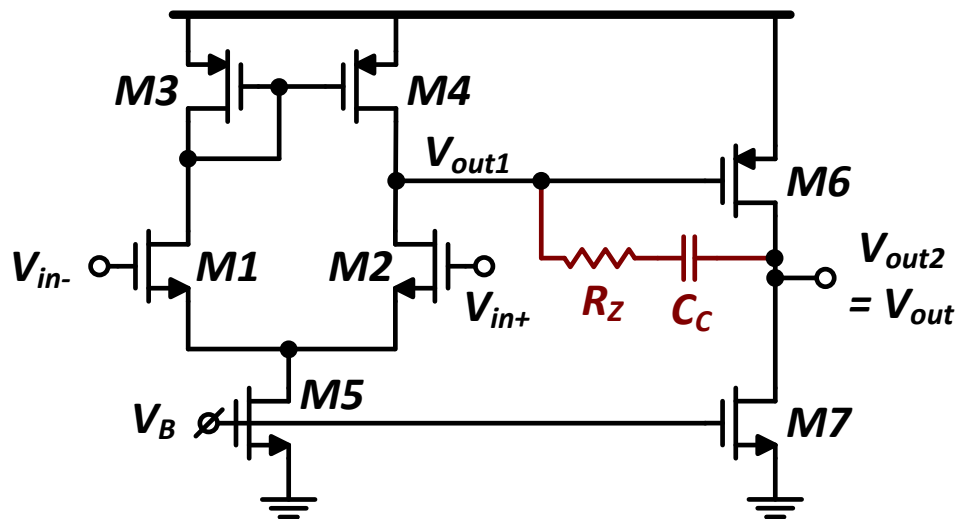


Miller Zero Placement

- Place the zero at ∞

$$\omega_z = \frac{1}{C_C \left(\frac{1}{g_{m6}} - R_Z \right)}$$

$$R_Z = \frac{1}{g_{m6}}$$

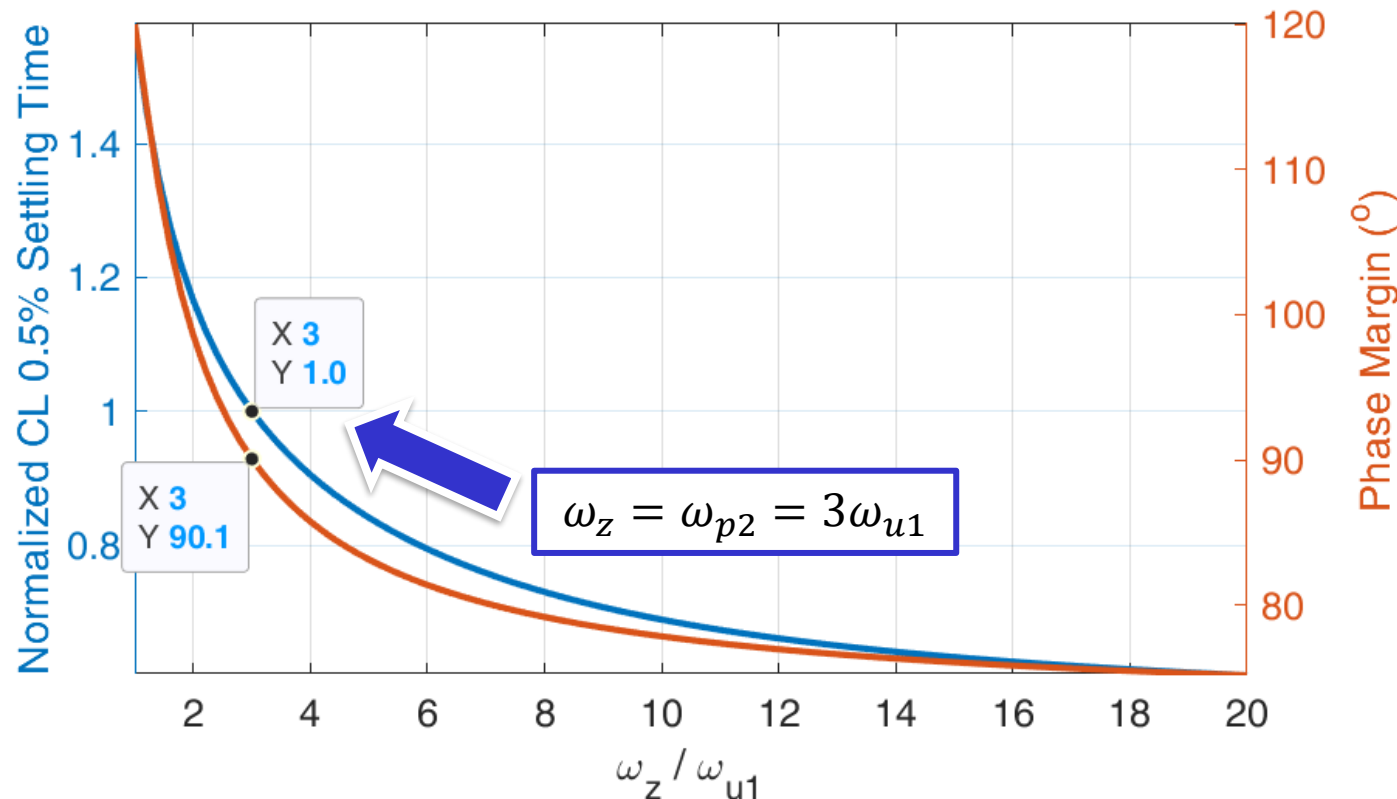


Miller Zero Placement

- ❑ Some designers try to move the zero to the LHP to cancel ω_{p2} and improve the PM

$$s_Z = \frac{1}{C_C \left(\frac{1}{g_{m6}} - R_Z \right)} \approx -\frac{g_{m6}}{C_1 + C_2} \rightarrow R_Z = \frac{C_1 + C_2 + C_C}{g_{m6} C_C}$$

- Practically never achieved and actually leads to slower response!

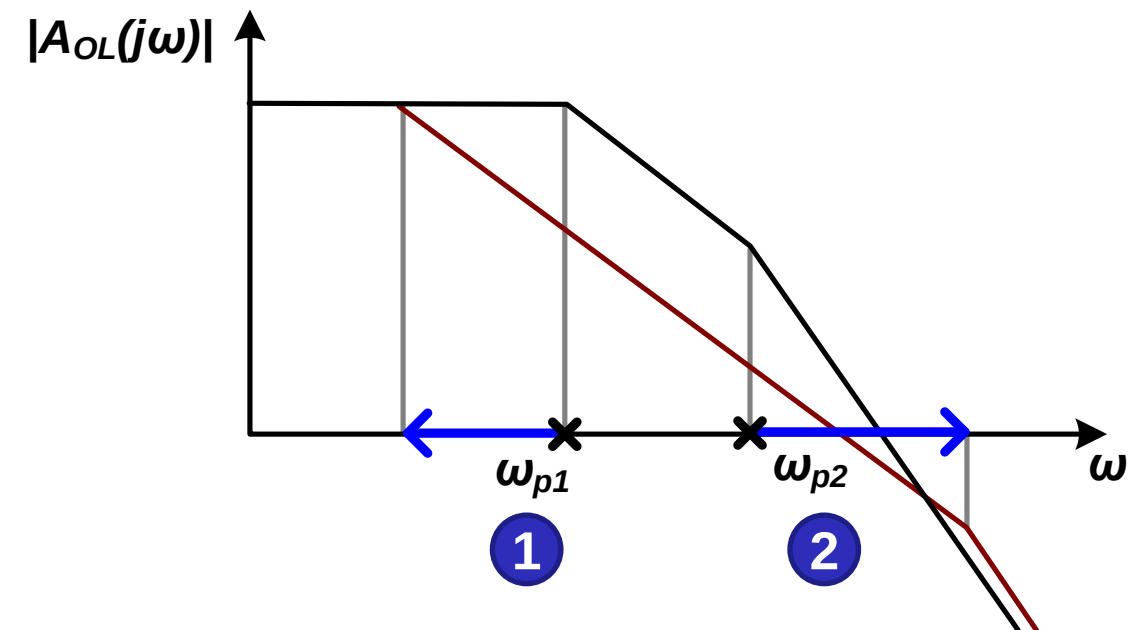
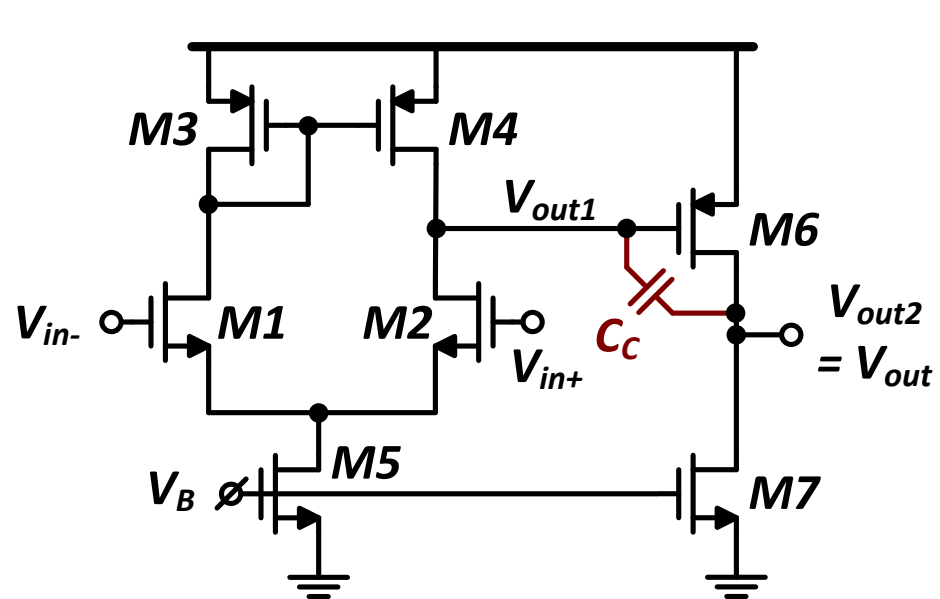


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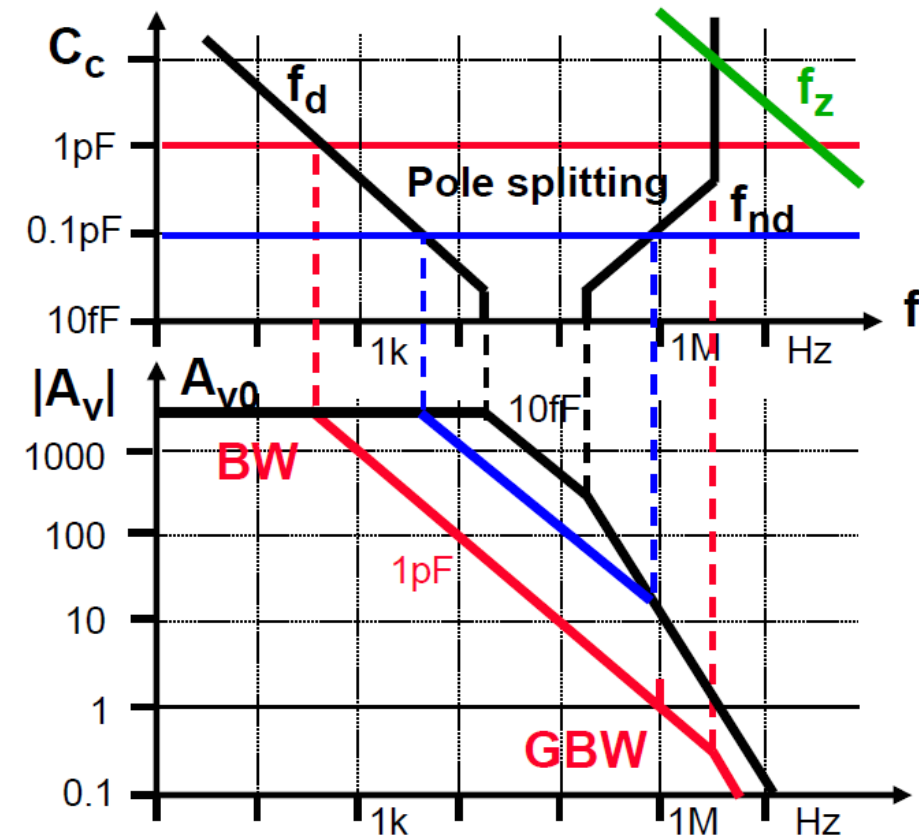
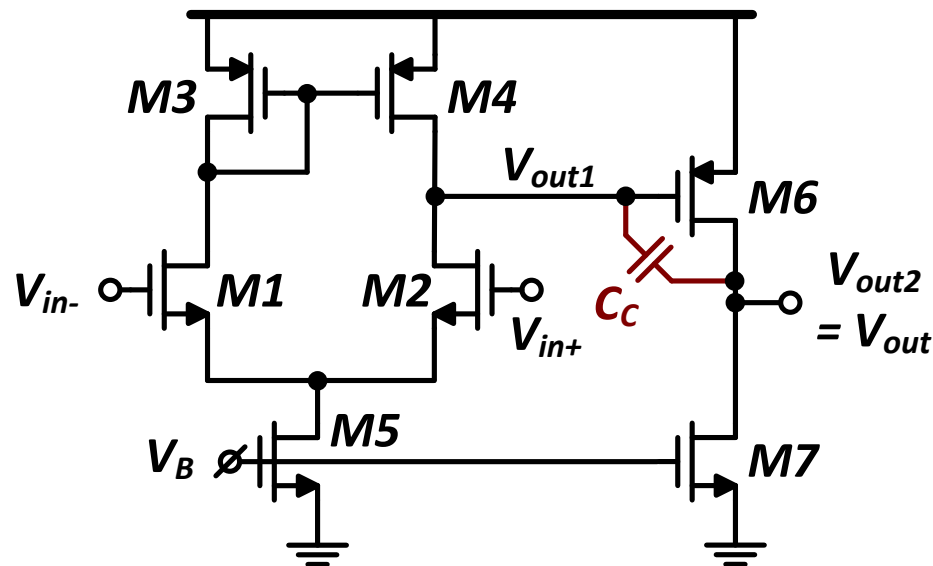
Miller Compensation: Pole Splitting

	ω_{p1}	ω_{p2}
Before	$\frac{1}{R_{out1}C_1}$	$\frac{1}{R_{out2}C_2}$
After	$\frac{1}{R_{out1}(\textcolor{red}{G}_{m2}R_{out2})\textcolor{green}{C}_C}$	$\frac{\textcolor{red}{G}_{m2}}{C_1 + C_2}$



Selecting C_C

- ❑ Too large C_C does not give more pole splitting: just smaller GBW
 - Usually we choose $C_1 < C_C < C_L$
 - Reasonable starting point: $C_C \approx (0.3 \rightarrow 0.5) \times C_L$



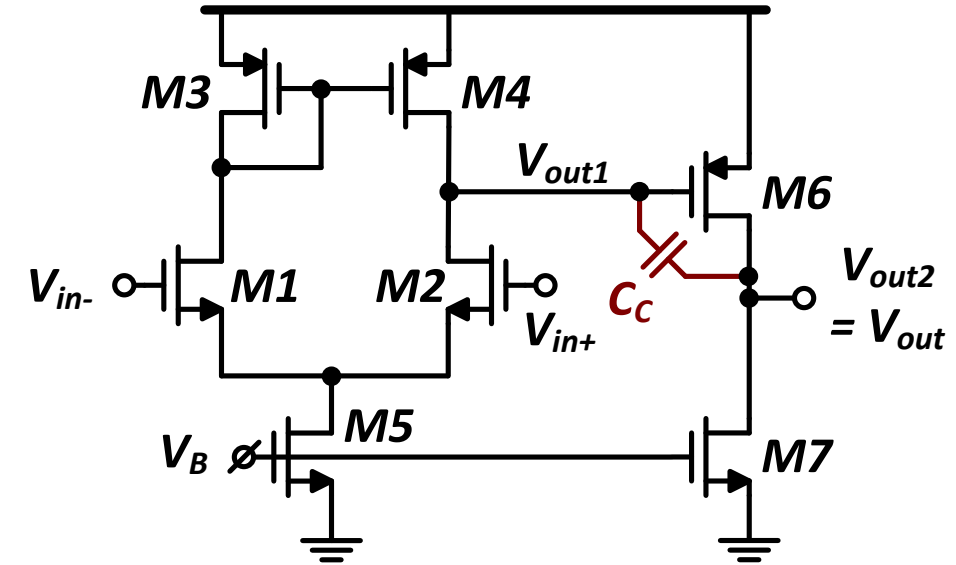
Stability and PM: ω_{p2}/ω_{u1}

$$\omega_{p1} \approx \frac{1}{R_{out1}(G_{m2}R_{out2})C_C}$$

$$\omega_{p2} \approx \frac{G_{m2}}{C_1 + C_2} \approx \frac{G_{m2}}{C_L}$$

$$GBW \approx G_{m1}R_{out1}G_{m2}R_{out2} \cdot \frac{1}{R_{out1}(G_{m2}R_{out2})C_C}$$

$$GBW = \omega_{u1} \approx \frac{G_{m1}}{C_C}$$



□ For critically damped response: $\zeta = 1$,

$Q = 0.5$, and $PM \approx 76^\circ$

$$\frac{\omega_{p2}}{\omega_{u1}} \approx 4 \rightarrow \frac{G_{m2}}{G_{m1}} \approx \frac{4C_L}{C_C}$$

Bias Currents in 1st and 2nd Stages

$$\frac{G_{m2}}{G_{m1}} \approx \frac{4C_L}{C_C}$$

- Assume $C_L = 2\text{pF}$ and $C_C = C_L/2 = 1\text{pF}$

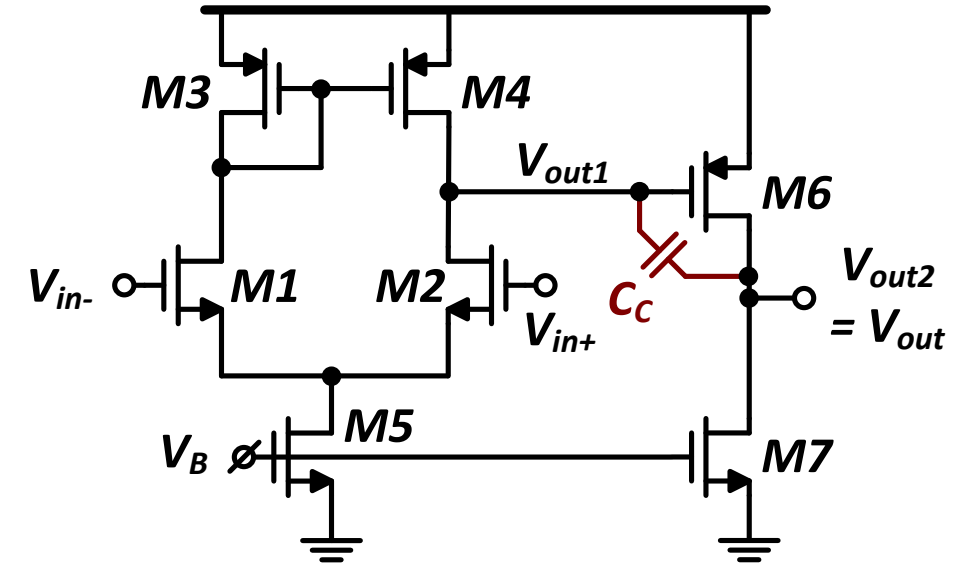
$$\frac{G_{m2}}{G_{m1}} = 4 \times \frac{2}{1} = 8$$

- If both stages use the same gm/ID

- Note that $I_{B1} = 2 \times I_{D1,2}$

$$\frac{I_{B2}}{I_{B1}} = 4$$

- 80% of the power is consumed in the second stage to achieve stability
 - 80% of bias current do not contribute (directly) to GBW
 - Miller OTA is very energy inefficient!
 - But it is still very robust and very popular!



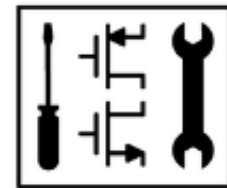
Outline

- ❑ The fundamentals
 - Feedback and stability
 - Frequency compensation and the optimal phase margin
 - Why two-stage amplifier?
- ❑ The two-stage amplifier
 - Miller compensation
 - The feedforward zero
 - Design guidelines
- ❑ The practice
 - Behavioral simulations
 - Design two-stage amplifier in less than 2 minutes!

ADT Analog Design Challenge

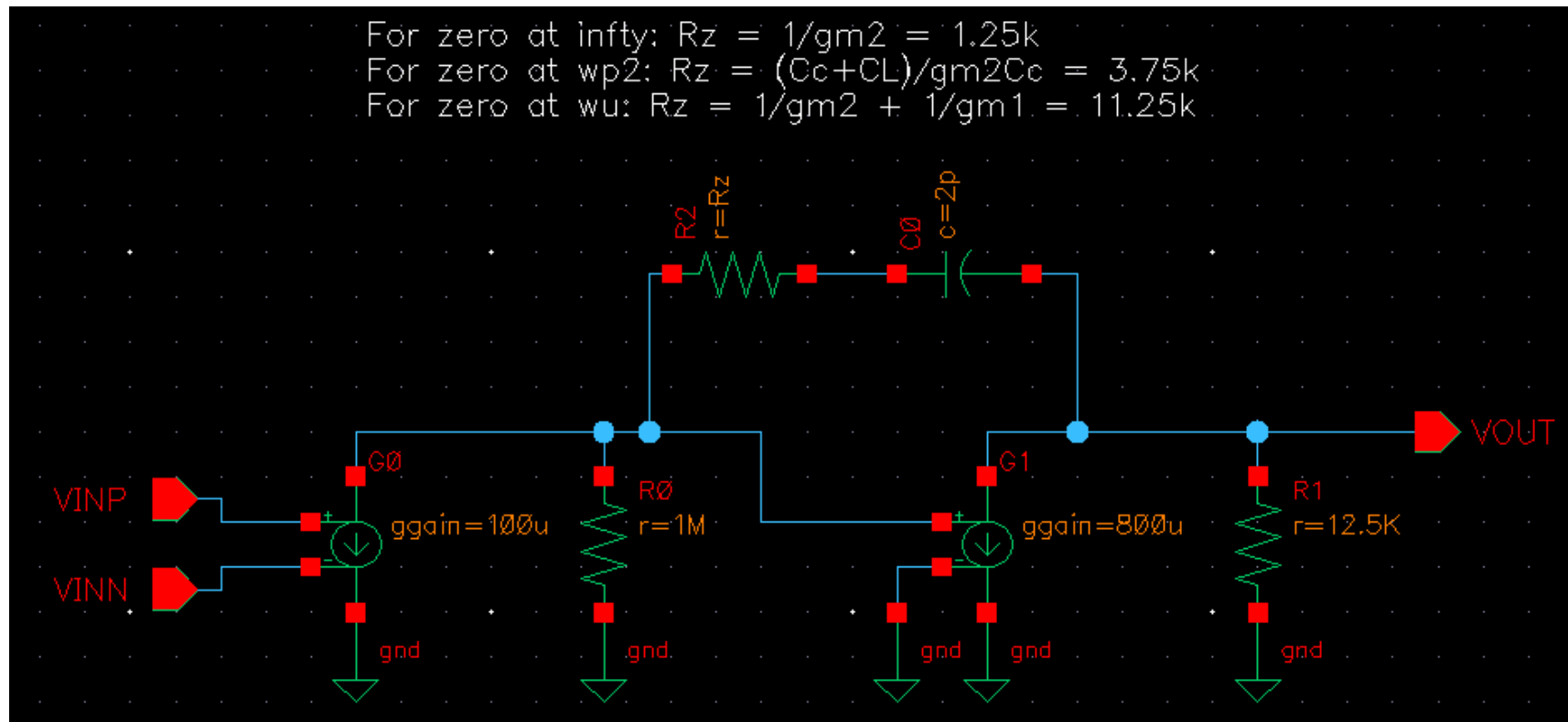


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Amazon Vouchers
and \$1000 ADT
Licenses!**



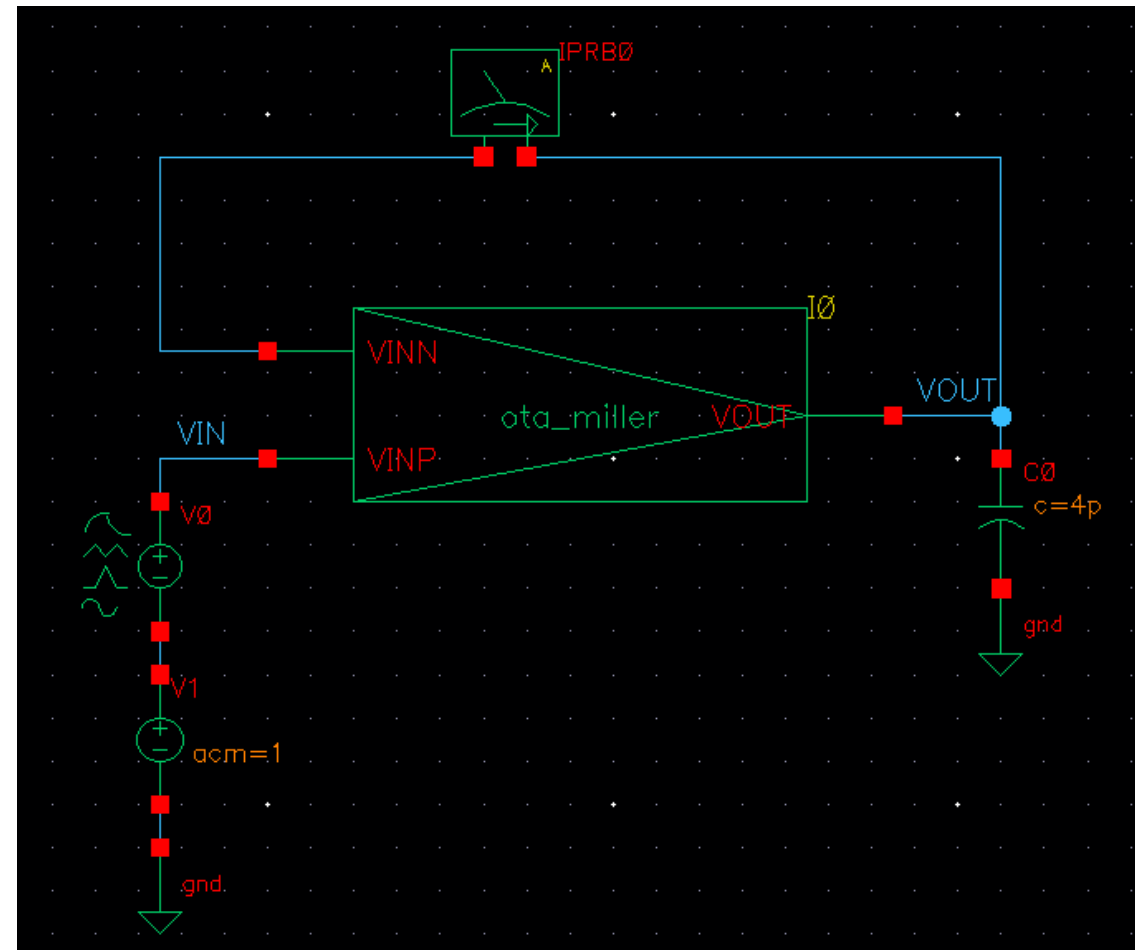
Miller OTA Behavioral Model

- ❑ Compare three different strategies for Miller zero placement
 - ∞
 - Cancel ω_{p2}
 - At ω_u (this is a designer mistake rather than a valid strategy!)

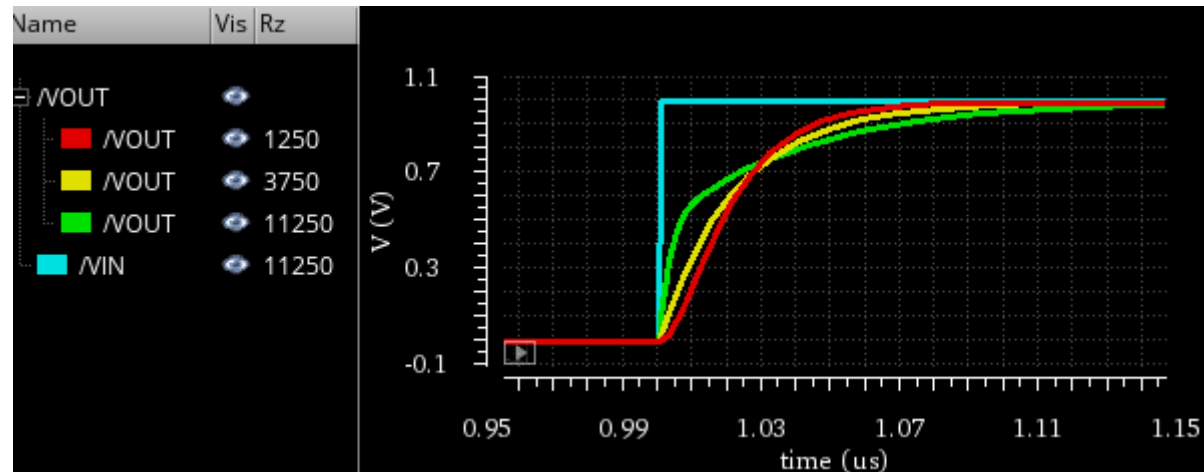
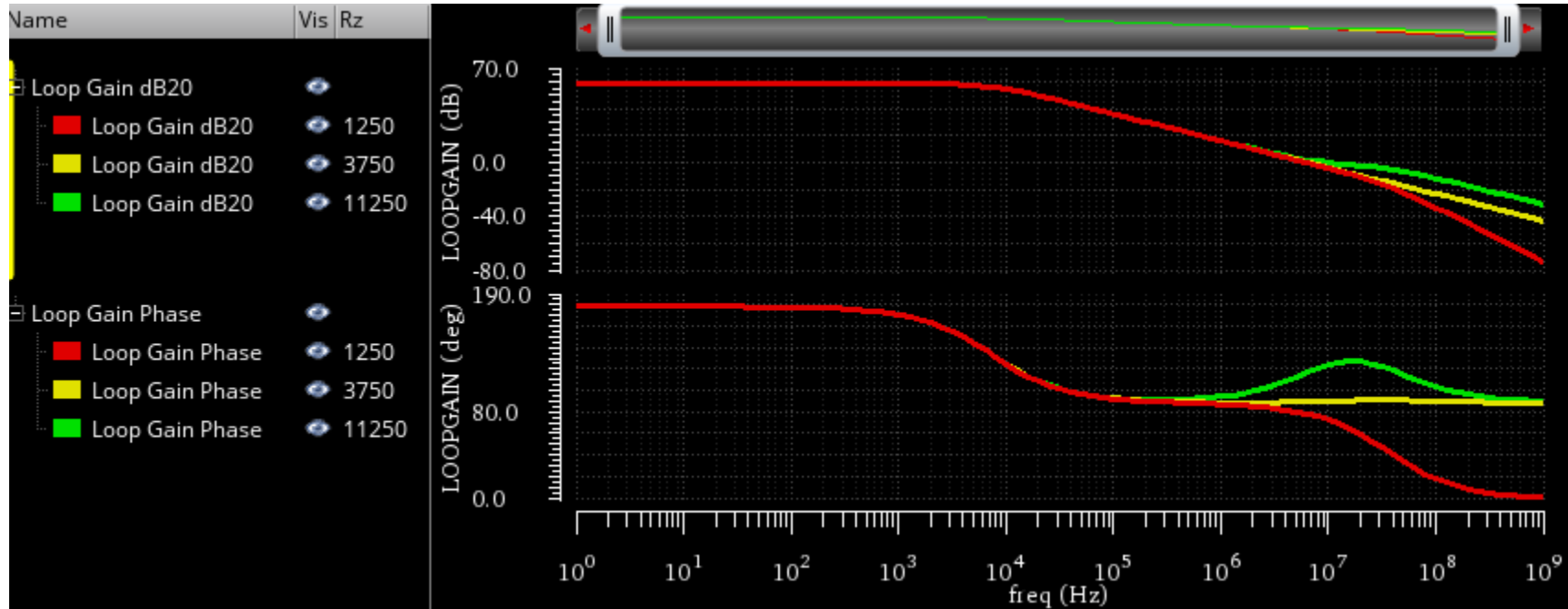


TB for LG and Step Response

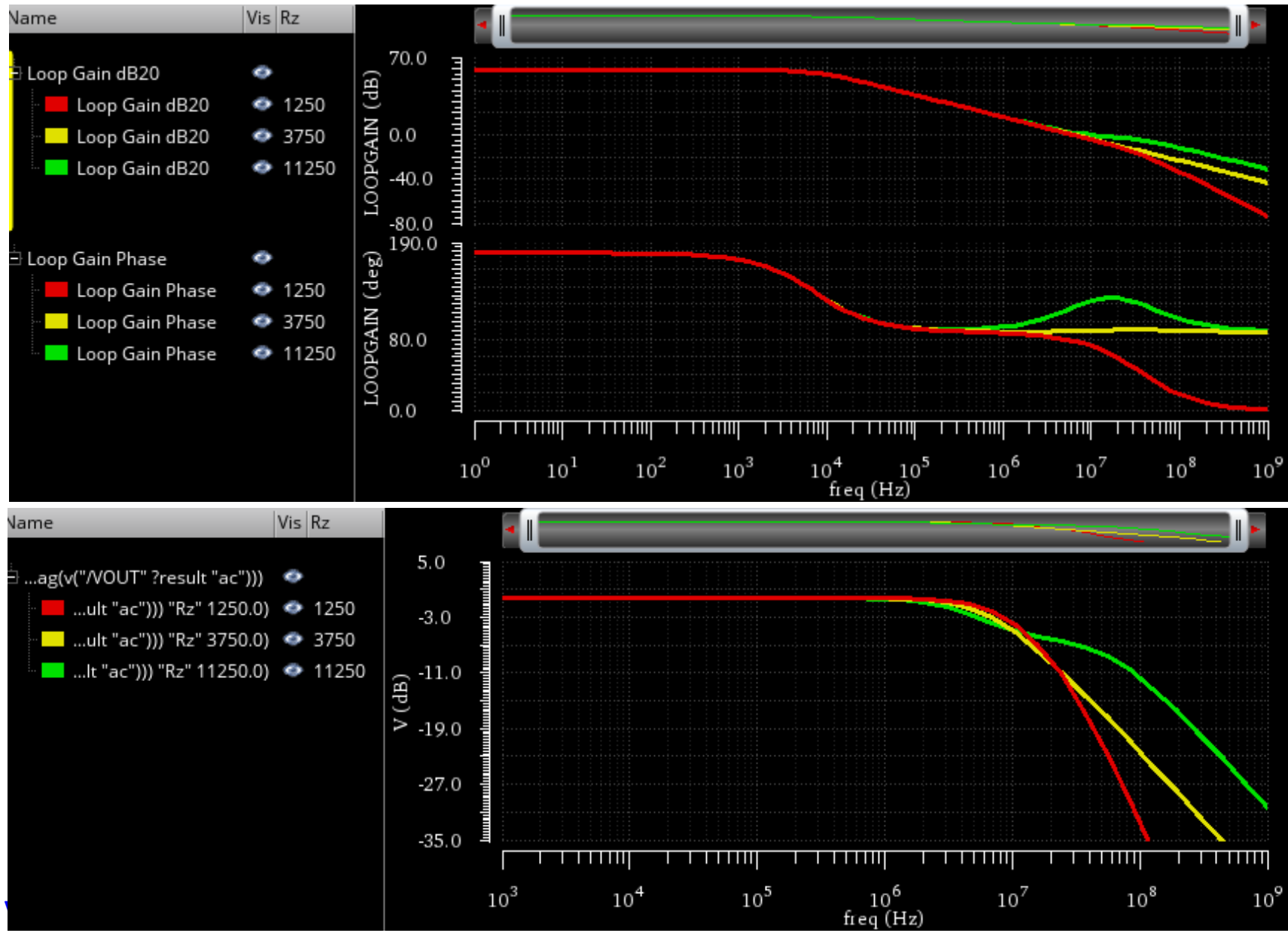
- Verify stability, ac response, and transient response in unity-gain buffer configuration.



LG and Step Response



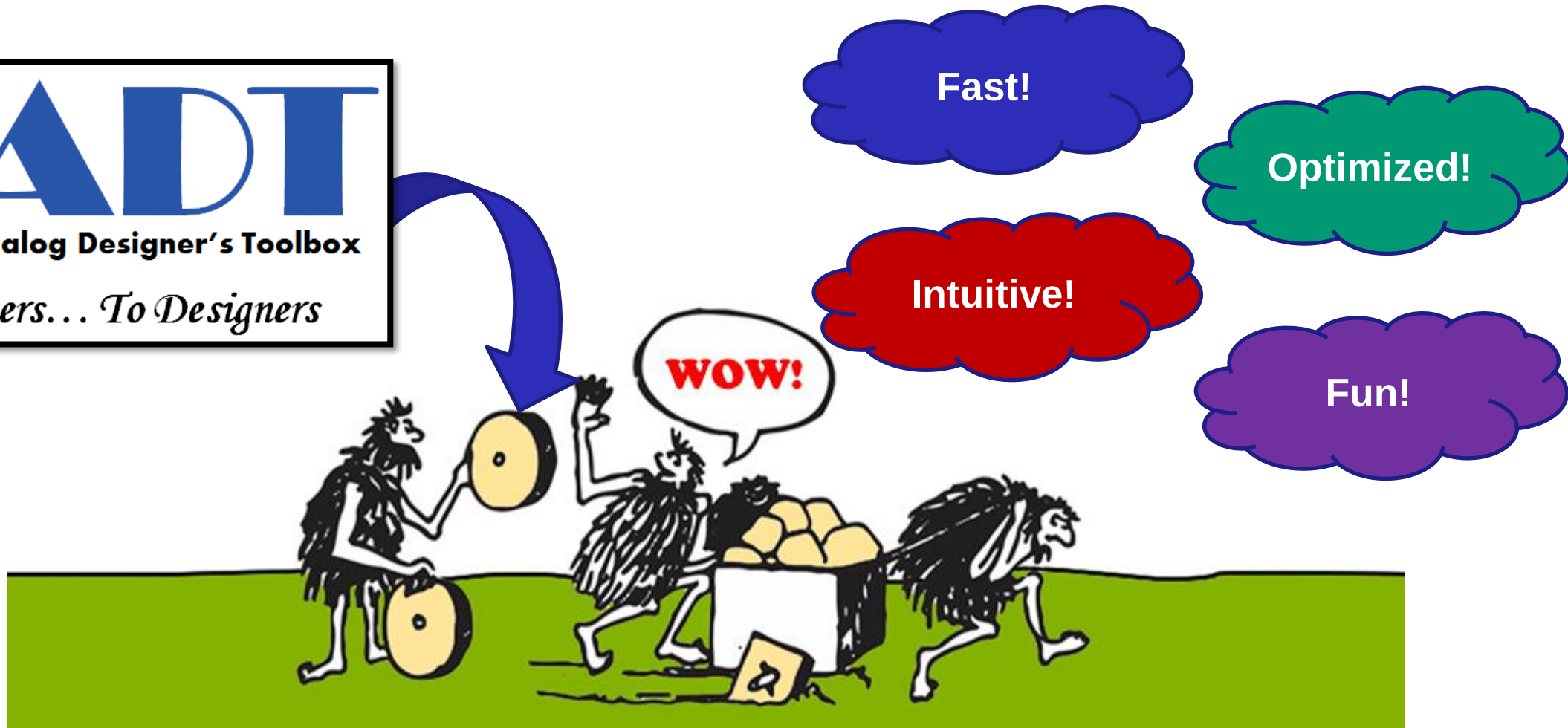
LG and CL Gain



Outline

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ADT: A Paradigm Change!



"If you can't describe what you are doing as a process, you don't know what you're doing."

W. Edwards Deming

References

- ❑ A. Sedra and K. Smith, “Microelectronic Circuits,” Oxford University Press, 7th ed., 2015
- ❑ B. Razavi, “Design of Analog CMOS Integrated Circuits,” McGraw-Hill, 2nd ed., 2017.
- ❑ W. Sansen, “Analog design essentials,” Springer, 2006.
- ❑ T. C. Carusone, D. Johns, and K. W. Martin, “Analog Integrated Circuit Design,” 2nd ed., Wiley, 2012.
- ❑ B. Murmann, EE214 Course Reader, Stanford University.

So, What's Next?

- ❑ Download ADT NOW and get your free license!
 - adt.master-micro.com
- ❑ Start using ADT and send us your feedback!
 - Enhancements, feature requests, etc.!
 - Your feedback really matters!
 - **We want to make designers happy 😊**

Thank you!

The Funniest ADT User Feedback 😊

ADT is like a "Magic Hat" which seems like an ordinary hat, but when the magician puts his hand in it, he gets a rabbit and other things that amaze everyone.

ADT looks very simple with a simple GUI, but when I started to use it, I found many rabbits every time I used it. I was amazed.



V-star (V^*)

- V-star (V^*) is inspired by V_{ov} but calculated from actual simulation data

$$g_m = \frac{2I_D}{V^*} \leftrightarrow V^* = \frac{2I_D}{g_m} = \frac{2}{g_m/I_D}$$

- Figures-of-merit in terms of V^*

$$g_m r_o = \frac{2I_D}{V^*} \cdot \frac{1}{\lambda I_D} = \frac{2}{\lambda V^*}$$

$$f_T = \frac{g_m}{2\pi C_{gg}} = \frac{1}{2\pi} \cdot \frac{2I_D}{V^*} \cdot \frac{1}{C_{gg}}$$

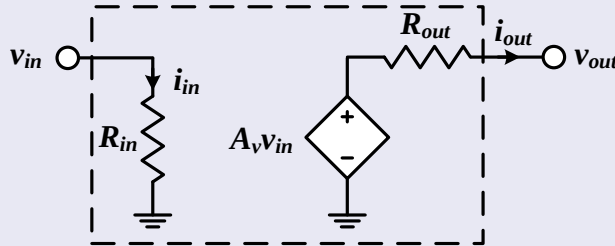
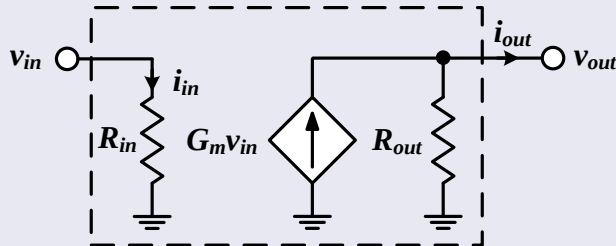
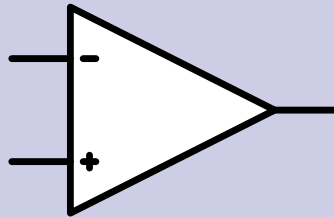
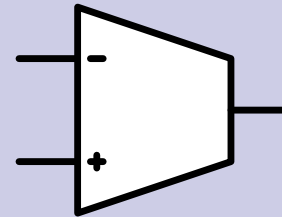
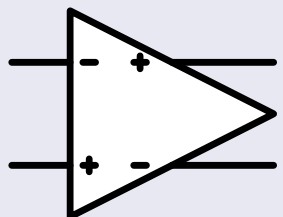
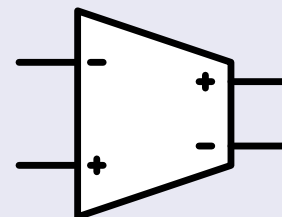
$$\frac{g_m}{I_D} = \frac{2}{V^*}$$

- The boundary between weak and strong inversion ($n \approx 1.2 \rightarrow 1.5$)

$$V_{ov}(SI) = V^*(WI) = 2nV_T \approx 60 \rightarrow 80mV$$

Op-Amp vs OTA

- ❑ An operational transconductance amplifier (OTA) is an op-amp without an output buffer
- ❑ Some designers just use op-amp name and symbol for both

	Op-amp	OTA
Rout	LOW	HIGH
Model		
Diff input, SE output		
Fully diff		

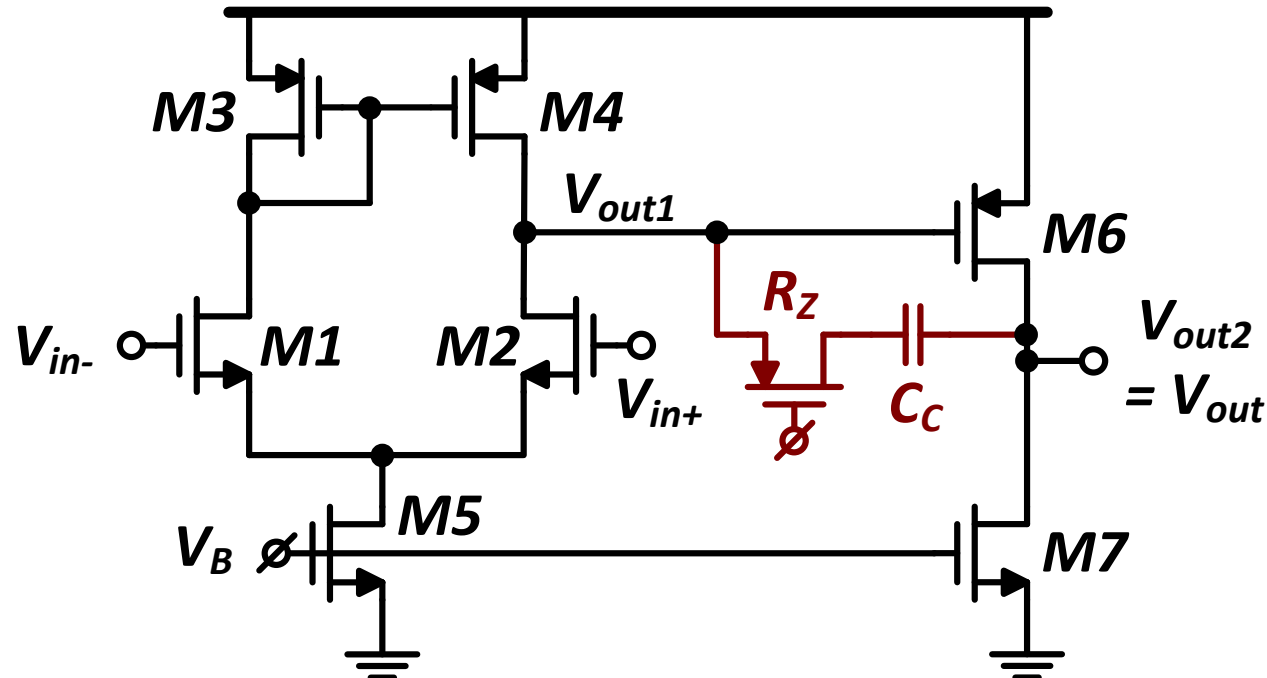
Rough Hand Analysis Example

- ❑ You always need to do rough hand analysis!
 - Get a sense of reasonable values and design trade-offs!
 - Define reasonable search space limits in ADT!

CL	Given	1.00E-12
GBW	Given	1E+08
wp2/wu1	Assume	2
(Gm/ID)1	Assume	10
(Gm/ID)2	Assume	10
CC	= CL/2	5.00E-13
Gm1	= 2*pi*CC*GBW	3.14E-04
Gm2	= wp2/wu1*CL/CC*Gm1	1.26E-03
I1	= 2*Gm1/(Gm/ID)1	6.28E-05
I2	= Gm2/(Gm/ID)2	1.26E-04
Ibtot	= I1+I2	1.88E-04

R_Z Implementation

- R_Z can be implemented as a simple resistor or using a transistor in the linear region
 - Clever biasing can make it track variations (maintain $R_Z \approx \frac{1}{g_{m6}}$)
 - Why R_Z is implemented as PMOS?



Single vs Two-Stage OTA: Sensitivity to C_L

